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Improving Bounds on Hales–Jewett Numbers: Symmetric Colorings, SAT Solvers, Line-Family Variants, and Forcing Structures

Master Thesis

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Abstract

The Hales–Jewett theorem states that for all $t, r \geq 1$ there is an n such that every r -colouring of the grid $[t]^n$ contains a monochromatic combinatorial line; the least such n is the Hales–Jewett number $\text{HJ}(t, r)$. Exact values are known only for $t \leq 2$ and at $\text{HJ}(3, 2) = 4$; this thesis develops lower-bound methods for the general case. Colourings of $[t]^n$ invariant under all coordinate permutations factor through the type map onto a simplex of polynomial size, on which line-avoidance is a finite check on corner tuples; colourings reading a single integer-weighted letter count — *one-weight* colourings — collapse the problem one dimension further, onto $\mathbb{Z}$, where line-freeness becomes avoidance of homothets $b + kS$ of the weight set. An elementary radix argument shows that in every fixed dimension the one-weight colourings are exactly the symmetric ones, so the symmetric lower-bound problem is precisely this one-dimensional homothety problem. From it we derive the closed-form bound $\text{HJ}(t, r) \geq \lceil (G_r(S) - 1)/D_S \rceil$ in terms of Gallai homothety numbers, with van der Waerden’s classical bound as the arithmetic case, and compute the new values $G_3(\{0, 1, 3\}) = 42$, $G_3(\{0, 1, 4\}) = 57$, $G_3(\{0, 2, 5\}) \geq 77$, $G_2(\{0, 2, 3, 5\}) = 67$ and $G_2(\{0, 1, 5, 6\}) = 80$. The main quantitative results are the lower bounds $\text{HJ}(3, 3) \geq 22$ and $\text{HJ}(4, 2) \geq 14$, each realised by an explicit single weight and improving the previous bounds of 14 and 12. For the restricted line families we determine the sum-type bracket ceilings $\kappa_{\text{sum}}(3, 3) = 11$, $\kappa_{\text{sum}}(4, 2) = 10$ and $\kappa_{\text{sum}}(4, 3) = 96$, exhibit periodic one-weight palettes giving $\text{HJ}^{[12]}(3, 3) = \text{HJ}^{[12]}(4, 2) = \infty$, prove that symmetric colourings are blind to the interval restriction, assemble from the literature the exact interval ceiling $\lambda(3, r)$ for every r , and compute the exact value $\text{HJ}^{(1)}(3) = 5$. The remaining chapters analyse the boundary of the $(4, 2)$ record, which decomposes as a bracket-extremal palette plus a single boundary scale; determine the cyclic analogues over $\mathbb{Z}_t^n$, which are logarithmic in r ; and identify the symmetric problem with the weak chromatic threshold of a corner hypergraph, giving $\text{HJ}_{\text{sym}}(t, r) = \Omega(r^{t-1}/t)$ and two-sided quasipolynomial bounds on $\text{HJ}_{\text{sym}}(3, r)$. Every avoidance certificate is re-verified by direct enumeration independently of any solver; every forcing claim is cross-checked with independent SAT solvers.

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Notation

Throughout, $t \geq 2$ is the alphabet size, $r \geq 2$ the number of colours, and $n \geq 1$ the dimension. Each item below is fixed once and used without further comment.

Words and lines

$[t] = \{1, \dots, t\}$	the alphabet.
$[t]^n$	the <i>grid</i> : the set of words $w = w_1 \cdots w_n$ with letters in $[t]$ (elsewhere also called the combinatorial cube).
τ	a <i>root</i> : an element of $([t] \cup \{*\})^n \setminus [t]^n$, i.e. a word over $[t] \cup \{*\}$ using the symbol $*$ at least once.
$\tau(a)$	the word obtained from τ by replacing every $*$ by the letter $a \in [t]$.
$I(\tau) \subseteq [n]$	the <i>active set</i> of τ : the coordinates carrying $*$.
$L_\tau = \{\tau(1), \dots, \tau(t)\}$	the <i>combinatorial line</i> generated by τ ; over $[4]$, the root $2*1*$ generates $\{2111, 2212, 2313, 2414\}$.
$L([t]^n)$	the set of all combinatorial lines of $[t]^n$.
$L_{*...*}$	the <i>diagonal</i> : the line whose root is all-active.
$L^{[K]}([t]^n)$	the lines with at most K active coordinates.
$L^{(q)}([t]^n)$	the <i>q-fold lines</i> : those whose active set is a union of at most q subintervals of $[n]$.

$\sigma(w) = \sum_{i=1}^n w_i$ the *weight* (letter sum) of the word w .

Types, weights, and palettes

$\text{type}(w) = (a_1, \dots, a_t)$ the letter counts $a_j = |\{i : w_i = j\}|$ of the word w .

$T_n^{(t)}$ the discrete simplex $\{a \in \mathbb{Z}_{\geq 0}^t : \sum_j a_j = n\}$ of types, of size $\binom{n+t-1}{t-1}$; for $t = 3$ we abbreviate $T_n := T_n^{(3)}$.

e_a the a -th standard basis vector of $\mathbb{Z}^t$.

$C_{k,v} = \{v + ke_1, \dots, v + ke_t\}$ the *corner tuple* of scale $k \geq 1$ and base $v \in T_{n-k}^{(t)}$ (Chapter 3).

$\omega \in \mathbb{Z}^t$ a *weight*; $S_\omega = \{\omega_1, \dots, \omega_t\}$ is its value set.

$\langle \omega, a \rangle = \sum_j \omega_j a_j$ the weight functional on types.

D_ω, D_S the *diameter* $\max - \min$ of a weight ω , or of a finite set $S \subset \mathbb{Z}$.

χ a *palette*: a map $\mathbb{Z} \rightarrow [r]$, taken m -periodic when so indicated.

$c_{\omega, \chi}$ the *one-weight colouring*
 $c_{\omega, \chi}(w) = \chi(\langle \omega, \text{type}(w) \rangle)$; *sum-type* colourings are the case $\omega = (1, 2, \dots, t)$, where $\langle \omega, \text{type}(w) \rangle = \sigma(w)$.

$b + kS$ the *scale- k homothet* of a finite set $S \subset \mathbb{Z}$ based at b (always with $k \geq 1$).

$\langle S \rangle$ the additive monoid generated by S : all finite sums of elements of S .

$G_r(S)$ the *Gallai homothety number*: the least N such that every r -colouring of N consecutive integers contains a monochromatic homothet of S lying inside them (Chapter 2).

$R_r(E)$ the *Rado number* of a linear equation E : the least N such that every r -colouring of $\{1, \dots, N\}$ contains a monochromatic injective solution of E .

$\mathbb{Z}_m$

the integers modulo m .

Hales–Jewett numbers and their variants

$\text{HJ}(t, r)$	the least n such that every r -colouring of $[t]^n$ contains a monochromatic combinatorial line; $\text{HJ}(t) := \text{HJ}(t, 2)$.
$W(k, r)$	the van der Waerden number: the least N such that every r -colouring of $\{1, \dots, N\}$ contains a monochromatic k -term arithmetic progression.
C	a colouring class, $C \in \{\text{sum}, \text{ow}, \text{sym}, \text{all}\}$: the sum-type, one-weight, symmetric, and unrestricted classes of Section 2.2.
$\text{HJ}_C(t, r)$	the least n such that every C -colouring of $[t]^n$ has a monochromatic line; $\text{HJ}_{\text{all}} = \text{HJ}$.
$n_C(t, r)$	$= \text{HJ}_C(t, r) - 1$, the largest line-free dimension of the class; $n_\omega(t, r)$ is the largest line-free dimension of the single weight ω , its <i>reach</i> (Chapter 2).
$\text{HJ}_C^{[K]}(t, r)$	the least n such that every C -colouring of $[t]^n$ has a monochromatic line of $L^{[K]}([t]^n)$, and ∞ if no such n exists; the subscript is dropped for $C = \text{all}$.
$\text{HJ}_C^{(q)}(t, r)$	the same over the q -fold lines $L^{(q)}([t]^n)$.
$\kappa_C(t, r)$	the <i>bracket ceiling</i> $\max\{K \geq 0 : \text{HJ}_C^{[K]}(t, r) = \infty\}$.
$\lambda_C(t, r)$	the <i>interval ceiling</i> $\max\{q \geq 0 : \text{HJ}_C^{(q)}(t, r) = \infty\}$.
$\text{HJ}^{\text{cyc}}, \text{HJ}^{\text{unit}}, \text{HJ}^{\text{geom}}$	the cyclic, unit-cyclic, and geometric numbers over $\mathbb{Z}_t^n$ (Chapter 6).
$\text{ah}(t, n)$	the anti–Hales–Jewett threshold: the least r such that every surjective r -colouring of $[t]^n$ contains a rainbow line (Chapter 5).

Verification

Statements marked *computer-assisted* rest on finite computations. Avoidance certificates — palettes, simplex tables, interval colourings — are re-verified by direct enumeration from the definitions, independently of any solver; unsatisfiability (forcing) claims rest on SAT solvers and are marked with the number of independent solvers used. All certificates and code are available at <https://github.com/ysmouhib/hj-certificates>; the certificates themselves are reproduced in the Appendix.

An interactive companion, the *Hales–Jewett Explorer*, is available at <https://ysmouhib.github.io/hales-jewett-explorer/>, where the colourings can be explored directly.

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Chapter 1

Introduction

The Hales–Jewett theorem [1] states that for all $t, r \geq 1$ there is an n such that every r -colouring of the grid $[t]^n$ contains a monochromatic combinatorial line. The associated *Hales–Jewett numbers* $HJ(t, r)$ — the least such n — are known exactly only for $t \leq 2$ and at $(3, 2)$, where $HJ(3) = 4$ [2]. Between the lower bounds transferred from van der Waerden numbers and the upper bounds of Shelah [3], Lavrov [4], and Conlon [5] lies an enormous gap, and progress on either side requires understanding the line-free colourings at the frontier.

This thesis approaches the numbers through symmetric colorings and line-family variants: A colouring invariant under all coordinate permutations factors through the type map onto a simplex of polynomial size, on which line-avoidance is a finite, solver-independent check; a single integer weight collapses it one dimension further, onto $\mathbb{Z}$, where line-freeness becomes homothety-avoidance and van der Waerden’s and Gallai’s theorems apply directly. A short radix argument shows that the second reduction loses nothing: in every fixed dimension the one-weight colourings already realise the whole symmetric class. Both reductions are then exploited computationally, with SAT-produced certificates re-verified by direct enumeration.

The contributions, in order of presentation, are the following.

1. *A uniform lower-bound method, calibrated at small parameters* (Chapter 2). The weight functional turns the standard deduction of Gallai–Witt from Hales–Jewett around, yielding the closed-form bound $HJ(t, r) \geq \lceil (G_r(S) - 1)/D_S \rceil$ in terms of Gallai homothety numbers (Theorem 2.15), of which the van der Waerden bound is the arithmetic case. At $(3, 2)$ the bound is weight-independent and tight for *every* weight, by the complete determination of the two-colour homothety numbers of three-point sets (Proposition 2.21,

after Brown–Landman–Mishna and Kim–Rho); at $(3, 3)$ the new Gallai numbers $G_3(\{0, 1, 3\}) = 42$, $G_3(\{0, 1, 4\}) = 57$ and $G_3(\{0, 2, 5\}) \geq 77$ lift the closed form from the van der Waerden shadow 13 to $\text{HJ}(3, 3) \geq 16$, from a one-line certificate (Theorem 2.22).

2. *The identification of the symmetric class, and the records* (Chapter 3). An elementary radix argument identifies the one-weight colourings with the symmetric colourings in every fixed dimension (Theorem 3.5) — a simple observation with useful consequences: the apparent hierarchy of multi-weight constructions is flat, and the symmetric lower-bound problem is exactly a one-dimensional colouring problem. Its quantitative payoff is the main result of the thesis: single explicit weights give the records $\text{HJ}(3, 3) \geq 22$ (Theorem 3.7, via $\omega = (0, 1, 29)$) and $\text{HJ}(4, 2) \geq 14$ (Theorem 3.10, via $\omega = (0, 2, 3, 5)$, in closed form from the new Gallai numbers $G_2(\{0, 2, 3, 5\}) = 67$ and $G_2(\{0, 1, 5, 6\}) = 80$), improving the previous bounds $\text{HJ}(3, 3) \geq 14$ [6] and $\text{HJ}(4, 2) \geq 12$, so that $14 \leq \text{HJ}(4) < 10^{11}$; the records also appear in the companion note [7]. The same classical construction records $\text{HJ}(3, 4) \geq 38$, $\text{HJ}(5) \geq 45$ and $\text{HJ}(6) \geq 227$. At $(3, 3)$ the full ladder of certificates is charted — arithmetic closed form, 13; the triples $\{0, 1, 3\}$, $\{0, 1, 4\}$, 14; a 49-cell periodic palette, 15; the length-76 interval certificate, 16; the 253-cell radix table, 22.
3. *The bracket and interval variants* (Chapter 4). Restricting the line family probes the numbers along two axes. The bracket ceilings κ_C are determined at small parameters — $\kappa_{\text{sum}}(3, 3) = 11$, $\kappa_{\text{sum}}(4, 2) = 10$, $\kappa_{\text{sum}}(4, 3) = 96$ — and non-arithmetic periodic weights pass the van der Waerden ceiling, giving $\text{HJ}^{[12]}(3, 3) = \text{HJ}^{[12]}(4, 2) = \infty$. The transverse interval axis is provably invisible to the whole symmetric class (Proposition 4.15); there, assembling Shelah’s proof with the bounds of Conlon–Kamčev [8], Leader–Räty [9] and Kamčev–Spiegel [10] determines the interval ceiling for every r — $\lambda(3, r) = r - 1$ for odd and $r - 2$ for even r (Proposition 4.19) — and a SAT computation gives the first nontrivial exact interval value, $\text{HJ}^{(1)}(3) = 5$ (Theorem 4.20). Three conjectures — Collapse, Diagonal-only, and the extremality of symmetric colourings — organise the frontier.
4. *The complexity of the boundary, and a rainbow analogue* (Chapter 5). An anatomy of the $(4, 2)$ record palette locates the source of its compression: for scales up to twelve it is an extremal object of the bracket regime, valid at every base, and its reach decomposes as *bracket cap plus one boundary scale* (Proposition 5.3) — which also explains why no comparably compressed witness exists at $(3, 3)$. The unsatisfiable instance one dimension past the record is globally irreducible across resolution, width, and algebraic

measures, yet sits only three monochromatic copies from feasible. The anti-Ramsey companion improves the anti-Hales–Jewett bound to $\text{ah}(3, 4) \geq 25$ (Proposition 5.9).

5. *Cyclic, unit-cyclic, and geometric lines* (Chapter 6). Over $\mathbb{Z}_t^n$ the cyclic numbers are logarithmic in r , against the at-least-linear classical ones, and the four families separate completely at the first nontrivial point:

$$\text{HJ}^{\text{cyc}}(3, 2) = 2 < \text{HJ}^{\text{unit}}(3, 2) = \text{HJ}^{\text{geom}}(3, 2) = 3 < \text{HJ}(3, 2) = 4.$$

6. *Forcing structures and asymptotic bounds* (Chapter 7). On the upper-bound side, monochromatic lines are forced only by large, global substructures, never by a bounded clique or a Fano configuration. Quotienting by the coordinate action identifies the symmetric problem with the weak chromatic threshold of a linear *corner hypergraph*; this yields $\text{HJ}_{\text{sym}}(t, r) = \Omega(r^{t-1}/t)$ in two lines by the Local Lemma, and two-sided quasipolynomial bounds $e^{c \log^2 r} \leq \text{HJ}_{\text{sym}}(3, r) \leq e^{(\log r)^{O(1)}}$ via the corners theorem — making the extremality conjecture falsifiable on the asymptotic axis.

Chapter 8 collects the resulting open problems. Every computational claim below is verified in two independent ways: avoidance certificates — palettes, simplex tables, and interval colourings — are re-verified by direct enumeration from the definitions, independently of any solver, while unsatisfiability (forcing) claims rest on SAT solvers and are marked with the number of independent solvers used. The certificates are reproduced in the Appendix; the SAT encoders, standalone verifiers, and census data are collected in the public repository <https://github.com/ysmouhib/hj-certificates>.

An interactive companion, the *Hales–Jewett Explorer*, is available at <https://ysmouhib.github.io/hales-jewett-explorer/>, where the colourings can be explored directly.

Chapter 2

Classical Bounds: van der Waerden and Gallai–Witt

A lower bound $\text{HJ}(t, r) > n$ is a line-free r -colouring of $[t]^n$. This chapter develops the one-dimensional theory of such colourings: after fixing the theorem and the known values (Section 2.1) and the type map with its weight functionals (Section 2.2), we recall van der Waerden’s bound as the case of the arithmetic weight (Section 2.3) and generalise it at once, per the classical deduction of Gallai–Witt from Hales–Jewett run in reverse, to a closed-form bound in terms of Gallai homothety numbers (Section 2.4). The bound is then calibrated (Section 2.5): weight-independent and tight at $(3, 2)$, and lifted by new three-colour Gallai numbers to $\text{HJ}(3, 3) \geq 16$ from a one-line certificate.

2.1 The Theorem and the Known Values

Theorem 2.1 (Hales–Jewett [1]). *For all $t, r \geq 1$ there is an n such that every r -colouring of $[t]^n$ contains a monochromatic combinatorial line.*

The least such n is the Hales–Jewett number $\text{HJ}(t, r)$. The exact value is known only in a handful of cases.

Proposition 2.2. $\text{HJ}(2, r) = r$.

Proof. In $[2]^r$ let w_k have its first k coordinates equal to 1 and the rest to 2 ($0 \leq k \leq r$). For $a < b$ the words w_a, w_b form a combinatorial line (active on coordinates $a + 1, \dots, b$), so $w_0, \dots, w_r$ are pairwise collinear; any r -colouring of these $r + 1$ words repeats a colour on some pair, giving a monochromatic line,

whence $\text{HJ}(2, r) \leq r$. Conversely, colouring $w \in [2]^{r-1}$ by its number of 1's leaves every line bichromatic. $\square$

The first nontrivial value is $\text{HJ}(3) = 4$: Figure 2.1 shows a line-free 2-colouring of $[3]^3$ (from [11]), and Hindman and Tressler [2] proved $\text{HJ}(3) \leq 4$ by two independent arguments. For $t \geq 4$ no value of $\text{HJ}(t)$ is known. Lavrov [4] reduced the upper bound for $\text{HJ}(4)$ from Shelah's tower to $\text{HJ}(4) < 10^{11}$; with the lower bound 14 of this thesis (Theorem 3.10), $14 \leq \text{HJ}(4) < 10^{11}$. For three colours, Conlon [5] proved $\text{HJ}(3, r) \leq 2^{2^{cr}}$, so with Theorem 3.7 below, $22 \leq \text{HJ}(3, 3) \leq 2^{2^{3c}}$.

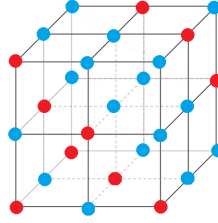


Figure 2.1: A line-free 2-colouring of $[3]^3$.

Remark 2.3 (Shelah's bound). Shelah's proof [3] of Theorem 2.1 (well exposed in Jungić [12], §5.2) replaces the Ackermann-type bound of the original double induction by a primitive-recursive one. The argument moreover yields monochromatic q -fold lines with $q \leq \text{HJ}(t-1, r)$ (Kamčev–Spiegel [10]), motivating the bracket and interval variants of Chapter 4.

2.2 The Type Map and One-Weight Colourings

The lower-bound constructions of this thesis all factor through the type map $w \mapsto \text{type}(w) \in T_n^{(t)}$, whose range is polynomial in n against the t^n words of the grid.

Lemma 2.4. *The S_n -orbits of $[t]^n$ under coordinate permutation are exactly the fibers of type. Consequently a colouring is symmetric (invariant under all coordinate permutations) if and only if it factors as $c = \bar{c} \circ \text{type}$ through a function $\bar{c}: T_n^{(t)} \rightarrow [r]$, which is then unique. We call c the lift of $\bar{c}$ and $\bar{c}$ the descent of c .*

2.2. The Type Map and One-Weight Colourings

Proof. A permutation preserves the multiset of letters, so type is constant on orbits; conversely $\text{type}(w) = \text{type}(w')$ makes the level sets $\{i : w_i = j\}$ equinumerous for each j , and any $\sigma \in S_n$ matching them satisfies $\sigma \cdot w = w'$. Thus symmetry is constancy on fibers, i.e. factorization through type; uniqueness holds because type is onto, realised by $1^{a_1} \dots t^{a_t}$. $\square$

The basic computation, used throughout, is the effect of a root on the type.

Lemma 2.5 (Type identity). *For a root τ with $k = |I(\tau)|$ active coordinates and inactive type $v \in T_{n-k}^{(t)}$,*

$$\text{type}(\tau(a)) = v + k e_a \quad (a \in [t]), \quad (2.1)$$

where e_a is the standard basis of $\mathbb{Z}^t$; and every pair (k, v) with $1 \leq k \leq n$ is realised, by $*^k 1^{v_1} \dots t^{v_t}$.

Proof. In $\tau(a)$ the k active coordinates carry a and the inactive ones contribute v . $\square$

Definition 2.6. A one-weight colouring of $[t]^n$ is a symmetric colouring

$$c_{\omega, \chi}(w) = \chi(\langle \omega, \text{type}(w) \rangle)$$

given by a weight $\omega \in \mathbb{Z}^t$ and a palette $\chi: \mathbb{Z} \rightarrow [r]$. The *sum-type* colourings are the case $\omega = (1, 2, \dots, t)$, where $\langle \omega, \text{type}(w) \rangle = \sigma(w)$ and $c = \chi \circ \sigma$.

Remark 2.7. Weights ω and $\alpha\omega + \beta\mathbf{1}$ ($\alpha \neq 0$) define the same one-weight family, since $\langle \alpha\omega + \beta\mathbf{1}, \text{type}(w) \rangle = \alpha\langle \omega, \text{type}(w) \rangle + \beta n$ is an affine reparametrization absorbed into χ . Thus ω matters only up to scaling and translation, and at $t = 2$ every nonconstant weight is affinely $(1, 2)$.

The classes stratify by how much of the type they read — the linear weight σ , an arbitrary weight, an arbitrary $\bar{c}$ —

$$\text{sum} \subseteq \text{ow} \subseteq \text{sym} \subsetneq \text{all},$$

the inner inclusions collapsing at $t = 2$ by Remark 2.7. For each class we attach a restricted Hales–Jewett number once, here, and reuse it throughout.

Definition 2.8. For a class C of colourings, $\text{HJ}_C(t, r)$ is the least n such that every C -colouring of $[t]^n$ has a monochromatic line ($\text{HJ}_{\text{all}} = \text{HJ}$). Shrinking C relaxes the quantifier, so

$$\text{HJ}_{\text{sum}} \leq \text{HJ}_{\text{ow}} \leq \text{HJ}_{\text{sym}} \leq \text{HJ}_{\text{all}} = \text{HJ}(t, r),$$

all finite. Superscripts (Chapter 4) restrict the *line* sought; the subscript C always means “quantify only over C ”.

2.3 Van der Waerden as the Arithmetic Weight

Theorem 2.9 (van der Waerden [13]). *For all k, r there is an n such that every r -colouring of $[n]$ contains a monochromatic k -term arithmetic progression; the least such n is $W(k, r)$.*

Proof. Set $N = \text{HJ}(k, r)$, $n = (k - 1)N + 1$. Given $c: [n] \rightarrow [r]$, define $c'': [k]^N \rightarrow [r]$ by $c''(w) = c(\sigma(w) - N + 1)$ (valid since $\sigma(w) \in [N, kN]$). By (2.1) a root with active set of size k' maps under σ to a k -term progression of gap k' , so the monochromatic line of $[k]^N$ guaranteed by N yields a monochromatic k -term progression in $[n]$. $\square$

Corollary 2.10. $\left\lceil \frac{W(t, r) - 1}{t - 1} \right\rceil \leq \text{HJ}(t, r)$.

Proof. The construction gives $W(t, r) \leq (t - 1)\text{HJ}(t, r) + 1$; rearrange. $\square$

This bound is governed by the van der Waerden numbers. Kozik–Shabanov [14] proved $W(k, r) \geq c r^{k-1}$, whence $W(t, 2) = \Omega(2^t)$ and $\text{HJ}(t) = \Omega(2^t/t)$. For prime p and $2 \leq r \leq p$, the Blankenship–Cummings–Taranchuk recurrence [15] gives $W(p + 1, r) > p^{r-1}2^p$, generalising Berlekamp's $W(p + 1, 2) > p \cdot 2^p$ [16]; with Corollary 2.10 this yields $\text{HJ}(p + 1, r) \geq p^{r-2}2^p$, in particular $\text{HJ}(p + 1) \geq 2^p$. Concretely, $W(3, 4) = 76$, $W(5, 2) = 178$ and $W(6, 2) = 1132$ [17, 18] give $\text{HJ}(3, 4) \geq 38$, $\text{HJ}(5) \geq 45$ and $\text{HJ}(6) \geq 227$.

The colouring behind Corollary 2.10 is sum-type: it reads only the arithmetic weight $\omega = (0, 1, \dots, t - 1)$. The next section shows that *every* weight carries such a bound, with van der Waerden's theorem as the arithmetic special case.

2.4 The Gallai–Witt Bound

Van der Waerden's theorem is the one-dimensional, arithmetic case of a theorem about patterns. Recall from the Notation that a *homothet* of a finite set $F \subset \mathbb{Z}^d$ is a set $b + kF$ with base $b \in \mathbb{Z}^d$ and integer scale $k \geq 1$.

Theorem 2.11 (Gallai–Witt [19, 20, 21]). *For every finite $F \subset \mathbb{Z}^d$ and every r there is an N such that every r -colouring of $\{1, \dots, N\}^d$ contains a monochromatic homothet of F .*

The standard proof specialises Hales–Jewett: enumerate $F = \{f_1, \dots, f_t\}$ and map words to sums, $\pi(w) = \sum_{i=1}^t f_{w_i}$. By the type identity (2.1) a line with k active coordinates and inactive part summing to b satisfies $\pi(\tau(a)) = b + k f_a$, so

π carries lines onto homothets of F ; composing an r -colouring of a large enough box with π (after a translation) therefore forces a monochromatic homothet once $n = \text{HJ}(t, r)$. For $d = 1$ and $F = S_\omega$ the map π is exactly the weight functional $\pi_\omega = \langle \omega, \text{type}(\cdot) \rangle$ of Section 2.2. The method of this section runs the substitution in reverse — a colouring of the integers avoiding the homothets of S_ω pulls back to a line-free colouring of the grid — and turns the pattern into a parameter to optimise. Three steps make this quantitative: a number measuring one-dimensional avoidance, an exact dictionary between lines and homothets, and the closed-form bound they produce together.

Definition 2.12. Let $S \subset \mathbb{Z}$ be finite with $|S| \geq 2$ and *diameter* $D_S = \max S - \min S$. The *Gallai homothety number* $G_r(S)$ is the least N such that every r -colouring of N consecutive integers contains a monochromatic homothet of S lying inside them.

Lemma 2.13 (Finiteness). $G_r(S) \leq W(D_S + 1, r) < \infty$.

Proof. Homothets translate, so assume $\min S = 0$. A window of $W(D_S + 1, r)$ consecutive integers contains a monochromatic progression $c, c + k, \dots, c + D_S k$, and this progression contains the homothet $c + kS$. $\square$

Lemma 2.14 (Line–pattern correspondence). Let $\omega \in \mathbb{Z}^t$ and $1 \leq K \leq n$. Under $c_{\omega, \chi}$, a line with k active coordinates and inactive type v carries the levels

$$\langle \omega, v \rangle + k S_\omega$$

— the scale- k homothet of the weight set based at $b = \langle \omega, v \rangle$ — and is monochromatic iff χ is constant on it. Hence $c_{\omega, \chi}$ has no monochromatic line in $L^{[K]}([t]^n)$ iff χ has no monochromatic homothet $b + kS_\omega$ with

$$1 \leq k \leq K \quad \text{and} \quad b \in \langle \omega, T_{n-k}^{(t)} \rangle.$$

Proof. By (2.1) the a -th word of the line has level $\langle \omega, v + k e_a \rangle = b + k \omega_a$, and a one-weight colouring is constant on level fibres; conversely every pair (k, v) with $k \leq K$ is realised (Lemma 2.5). $\square$

The bases realised in dimension n form the truncated monoid

$$\langle \omega, T_{n-k}^{(t)} \rangle = \left\{ \sum_a c_a \omega_a : c_a \in \mathbb{Z}_{\geq 0}, \sum_a c_a \leq n - k \right\}$$

— with $\min \omega = 0$, a subset of the interval $[0, D_\omega(n - k)]$ that need not fill it. For a lower bound the distinction can be ignored: avoidance at all bases of the interval is harder than at the realised ones, and avoidance at all bases is exactly what G_r measures.

Theorem 2.15 (Gallai-number lower bound). *For every $S \subset \mathbb{Z}$ with $|S| = t$,*

$$\text{HJ}(t, r) \geq \left\lceil \frac{G_r(S) - 1}{D_S} \right\rceil, \quad \text{hence} \quad \text{HJ}(t, r) \geq \max_{|S|=t} \left\lceil \frac{G_r(S) - 1}{D_S} \right\rceil.$$

Proof. Normalise $\min S = 0$, write $D = D_S$, and let ω have value set S . Then π_ω maps $[t]^n$ into the window $I_n = \{0, \dots, Dn\}$ and, by Lemma 2.14, every line onto a homothet of S inside I_n . If $Dn + 1 \leq G_r(S) - 1$, some r -colouring ξ of I_n contains no monochromatic homothet of S at all, and $\xi \circ \pi_\omega$ is line-free, so $\text{HJ}(t, r) > n$. The largest such n is $\lfloor (G_r(S) - 2)/D \rfloor = \lceil (G_r(S) - 1)/D \rceil - 1$. $\square$

Corollary 2.16 (The arithmetic weight is van der Waerden). $G_r(\{0, 1, \dots, t - 1\}) = W(t, r)$, and for the arithmetic weight the bound is exact: $n_{(0,1,\dots,t-1)}(t, r) = \lfloor (W(t, r) - 2)/(t - 1) \rfloor$, recovering Corollary 2.10.

Proof. Homothets of $\{0, \dots, t - 1\}$ are exactly the t -term progressions, so $G_r = W(t, r)$. Exactness holds because here the realised bases fill the whole interval, $\langle \omega, T_m^{(t)} \rangle = [0, (t - 1)m]$: by Lemma 2.14 the realised homothets in dimension n are then *all* t -term progressions inside $[0, (t - 1)n]$, and such a window admits a progression-free r -colouring iff $(t - 1)n + 1 \leq W(t, r) - 1$. $\square$

Lemma 2.17 (Affine covariance). *For $c \geq 1$ and $d \in \mathbb{Z}$,*

$$G_r(cS + d) = c(G_r(S) - 1) + 1, \quad G_r(-S) = G_r(S).$$

Hence $\lceil (G_r(S) - 1)/D_S \rceil$ is invariant under $S \mapsto \pm cS + d$, and the maximum in Theorem 2.15 may be taken over S with $\min S = 0$ and $\gcd S = 1$. (For $|S| = 3$ the scaling identity is Theorem 1 of Brown–Landman–Mishna [22].)

Proof. Translation and reflection act on colourings. For scaling, split a window of N consecutive integers into its c residue classes: every homothet of cS lies in a single class, each class is order-isomorphic to a window of $\lceil N/c \rceil$ or $\lfloor N/c \rfloor$ consecutive integers under which the homothets of cS correspond to the homothets of S , and the classes are coloured independently. Avoidance at length N is therefore possible iff $\lceil N/c \rceil \leq G_r(S) - 1$. $\square$

Two readings of the correspondence close the section: the ceiling of a fixed weight, and the algebraic form of homothety-avoidance.

Proposition 2.18 (Weight ceiling). *Fix ω with $|S_\omega| = t$ and $\min \omega = 0$. The line-free dimensions of ω form a finite initial segment: $n_\omega(t, r) < \infty$. Over all n the realised scales in Lemma 2.14 exhaust $\mathbb{Z}_{\geq 1}$ and the realised bases exhaust the monoid $\langle S_\omega \rangle$, so the finiteness is precisely the case $d = 1$ of Theorem 2.11 with bases restricted to the monoid — a restriction that loses nothing.*

Proof. Padding the inactive type by the letter of weight 0 shows every homothet realised in dimension n is realised in dimension $n + 1$; hence the initial segment, and the exhaustion of scales and monoid bases over all n . For finiteness, write $S_\omega = gS'$ with $\gcd S' = 1$ and let N_0 be a bound past which $\langle S' \rangle$ contains every integer. Given any palette χ , van der Waerden applied to $x \mapsto \chi(gx)$ on the window $[N_0, N_0 + W(\max S' + 1, r) - 1]$ produces a monochromatic homothet $b' + kS'$ with $b' \geq N_0$; then $gb' + kS_\omega$ is monochromatic, its base lies in $\langle S_\omega \rangle$, and its base and scale are bounded by a constant $C(S, r)$ independent of χ . All homothets with base and scale below $C(S, r)$ are realised once $n \geq n_0(S, r)$, so no palette is line-free beyond n_0 . $\square$

Proposition 2.19 (Rado reading). *Let $S = \{s_1 < \dots < s_t\}$ and $\mathbf{s} = (s_1, \dots, s_t)$. There is a homogeneous linear system $E_{\mathbf{s}}$ of $t - 2$ independent equations in $x_1, \dots, x_t$, each with coefficient sum zero — hence partition regular by Rado’s criterion — whose integer solutions are exactly the tuples*

$$\mathbf{x} = a\mathbf{1} + d\mathbf{s}, \quad a, d \in \mathbb{Z}.$$

The solutions with pairwise distinct coordinates are those with $d \neq 0$; as sets, they are the homothets of S for $d \geq 1$ and of the reflected pattern $-S$ for $d \leq -1$. Thus “no monochromatic injective solution of $E_{\mathbf{s}}$ ” means “no monochromatic homothet of S or of $-S$ ”, and Lemmas 2.13 and 2.17 bound the Rado number of every such system.

Proof. The lattice $\Lambda_{\mathbf{s}} = \{c \in \mathbb{Z}^t : \sum_j c_j = \sum_j c_j s_j = 0\}$ of affine dependencies of $\mathbf{s}$ has rank $t - 2$; a $\mathbb{Z}$ -basis of it defines $E_{\mathbf{s}}$, and each basis vector satisfies $\sum_j c_j = 0$, which for a single equation is Rado’s column condition. The kernel of $E_{\mathbf{s}}$ contains $\text{span}_{\mathbb{Q}}(\mathbf{1}, \mathbf{s})$, and both have dimension 2 because the s_j are not all equal, so they coincide. Finally, from $\mathbf{x} = a\mathbf{1} + d\mathbf{s}$ we get $x_i - x_j = d(s_i - s_j)$: the coordinates are pairwise distinct iff $d \neq 0$, and then $\{x_1, \dots, x_t\} = a + dS$, a homothet of S or of $-S$ according to the sign of d . $\square$

Example 2.20. For $S = \{0, 1, 3\}$ the lattice is generated by $(2, -3, 1)$, so $E_{\mathbf{s}}$ is the single equation $z + 2x = 3y$. Its injective solutions are the triples $\{a, a + d, a + 3d\}$ with $d \neq 0$: the homothets of $\{0, 1, 3\}$ for $d \geq 1$ and of the reflected pattern $\{0, 2, 3\}$ for $d \leq -1$ (via $x = b + 3k, y = b + 2k, z = b$). Since the grid realises only positive scales (Lemma 2.14), the search for a line-free one-weight colouring at $\omega = (0, 1, 3)$ is the search for a window palette free of $\{0, 1, 3\}$ -homothets alone, whereas the *Rado number* of $z + 2x = 3y$ demands freeness from both patterns at once. The two thresholds are distinct; Remark 2.24 quantifies both at four colours.

2.5 Calibration: the Bound at Small Parameters

At $(2, r)$ the closed form is tight for *every* weight: all two-point sets are affinely $\{0, 1\}$ (Lemma 2.17) and $G_r(\{0, 1\}) = r + 1$ by pigeonhole, giving the exact value $\text{HJ}(2, r) = r$. The first nontrivial pair behaves the same way.

Proposition 2.21 (Flatness at $(3, 2)$; after Brown–Landman–Mishna [22] and Kim–Rho [23]). *Let $S = \{0, s, s + u\}$ with $\gcd(s, u) = 1$ and diameter $D = s + u$. Then*

$$G_2(S) = \begin{cases} 4D, & \{s, u\} = \{1, 4m\} \text{ for some } m \geq 1, \\ 4D + 1, & \text{otherwise,} \end{cases}$$

and consequently, for every three-point S , $\lceil (G_2(S) - 1)/D_S \rceil = 4 = \text{HJ}(3, 2)$: at $(3, 2)$ the closed form of Theorem 2.15 is weight-independent and tight.

Proof. Since a copy $b + k\{1, 1 + s, 1 + s + u\}$ equals $(b + k) + kS$, homothety-avoidance on an interval is the same for the two sets, so $G_2(S)$ is the number $f(s, u)$ of Brown–Landman–Mishna: the least N such that every 2-colouring of $[1, N]$ contains a monochromatic homothetic copy of $\{1, 1 + s, 1 + s + u\}$. They proved $f(s, u) \leq 4(s + u) + 1$, with equality in a large family of cases, leaving for the family $\{t', 4mt'\}$ two candidate values [22]; Kim and Rho settled it: $f(4mt', t') = f(t', 4mt') = 4(4mt' + t') - t' + 1$, and $f(s, u) = 4(s + u) + 1$ for every other pair [23, Theorem 12]. For primitive S the exceptional pairs force $t' = 1$, since $\gcd(t', 4mt') = t'$, giving the displayed dichotomy; general S reduces to this by Lemma 2.17. For the ratio, $(G_2(S) - 1)/D$ equals 4 in the generic case and $4 - 1/D$ in the exceptional one, so its ceiling is 4 in both, and $\text{HJ}(3, 2) = 4$ [2] makes the bound tight. As an independent check, all fourteen primitive classes with $D \leq 9$ were recomputed by SAT, the avoidance certificates re-verified by direct enumeration; every value matches the formula, the two exceptional classes in range being $G_2(\{0, 1, 5\}) = 20$ and $G_2(\{0, 1, 9\}) = 36$ ($m = 1, 2$). $\square$

For $t \geq 4$ or $r \geq 3$ no closed form for $G_r(S)$ is known and one computes by SAT, as in Dumitru–Prunescu [24] and here. Table 2.1 collects the values driving this thesis.

Theorem 2.22 ($\text{HJ}(3, 3) \geq 16$ in closed form; computer-assisted). $G_3(\{0, 1, 3\}) = 42$ and $G_3(\{0, 1, 4\}) = 57$ (avoidance certificates re-verified by direct enumeration; forcing confirmed by three independent solvers), and

$$G_3(\{0, 2, 5\}) \geq 77.$$

The latter is witnessed by the explicit 3-colouring ξ of $\{0, \dots, 75\}$

2.5. Calibration: the Bound at Small Parameters

Table 2.1: The Gallai-number bound from a single input. The values 42, 57, 67, 80 and the bound $G_3(\{0, 2, 5\}) \geq 77$ are new. Avoidance colourings at $G_r - 1$ (at 76 for $\{0, 2, 5\}$) were re-verified by direct enumeration (Appendix A.2); forcing at G_r was established by SAT — two independent solvers for 27, 35, 67, 80, three for 42 and 57. The rows $\{0, 2, 3, 5\}$ and $\{0, 1, 5, 6\}$ give the $(4, 2)$ record of Theorem 3.10; the rows at three colours give Theorem 2.22.

(t, r)	S	D	$G_r(S)$	$\lceil (G_r(S) - 1)/D \rceil$
$(2, r)$	$\{0, 1\}$	1	$r + 1$	r (tight)
$(3, 2)$	$\{0, 1, 2\}$	2	$9 = W(3, 2)$	4 (tight)
$(3, 3)$	$\{0, 1, 2\}$	2	$27 = W(3, 3)$	13
$(3, 3)$	$\{0, 1, 3\}$	3	42	14
$(3, 3)$	$\{0, 1, 4\}$	4	57	14
$(3, 3)$	$\{0, 2, 5\}$	5	≥ 77	$\geq \mathbf{16}$
$(4, 2)$	$\{0, 1, 2, 3\}$	3	$35 = W(4, 2)$	12
$(4, 2)$	$\{0, 2, 3, 5\}$	5	67	14
$(4, 2)$	$\{0, 1, 5, 6\}$	6	80	14

10201110020122122201020111002012212210
102011100201221222001020111002012212200

(listing $\xi(0), \dots, \xi(75)$), re-verified by direct enumeration to contain no monochromatic homothet of $\{0, 2, 5\}$. Since $\lceil (G_3(\{0, 2, 5\}) - 1)/5 \rceil \geq \lceil 76/5 \rceil = 16$, Theorem 2.15 gives

$$\text{HJ}(3, 3) \geq 16.$$

Remark 2.23 (Reading the new values). The closed form 16 exceeds both the arithmetic row (13, the van der Waerden shadow) and the previous published record $\text{HJ}(3, 3) \geq 14$ [6]: a one-line, hand-checkable certificate already beats the prior state of the art. At $r = 2$ the ratio is flat at 4 (Proposition 2.21); the computed values at $r = 3$ (13, 14, 14, ≥ 16 at $D = 2, 3, 4, 5$) leave its behaviour in the diameter open. The closed form counts *all* homothets in the interval, including those whose base lies outside the monoid $\langle S \rangle$, while the grid realises only the monoid bases; the dimension reach $n_\omega + 1$ therefore dominates the closed form, with equality for arithmetic S (Corollary 2.16) and for the $(4, 2)$ record — how far the reach can outrun the closed form is the subject of Chapter 3, where a radix weight carries it to 22. Read backwards, Theorem 2.15 gives $G_r(S) \leq D_S \text{HJ}(t, r) + 1$, sharp at $t = 2$ where $G_r(\{0, D\}) = rD + 1$ (Lemma 2.17), and the optimised bound becomes an extremal problem on Gallai numbers per unit diameter; by affine covariance the ratio sees only the affine class of S — $G_2(\{0, 2, 4, 6\}) = 2 \cdot 34 + 1 = 69$ exceeds 67, yet $\lceil 68/6 \rceil = 12$ merely reproduces the arithmetic row of Table 2.1.

Remark 2.24 (The weight $(0, 1, 3)$ at four colours, and a correction). By Example 2.20 and Lemma 2.14, a palette on the window $\{0, \dots, 3n\}$ free of monochromatic $\{0, 1, 3\}$ -homothets makes $c_{(0,1,3),\chi}$ line-free on $[3]^n$. An explicit 4-colouring of $\{0, \dots, 92\}$ with this property exists (Appendix A.2, verified exhaustively), so $G_4(\{0, 1, 3\}) \geq 94$; the resulting grid bound $HJ(3, 4) \geq \lfloor 92/3 \rfloor + 1 = 31$ is nonetheless weaker than the van der Waerden shadow $HJ(3, 4) \geq 38$ of Corollary 2.16 ($W(3, 4) = 76$): a window of length at least 114 would be needed to surpass it via this weight, and $\{0, 1, 3\}$ -avoidance is already at its satisfiability phase transition near length 93. The *Rado number* of $z + 2x = 3y$ is a different threshold: by Example 2.20 solution-freeness forbids the reflected pattern $\{0, 2, 3\}$ as well, which the window certificate does not (it contains monochromatic reflected copies). An explicit 4-colouring of 56 consecutive integers free of monochromatic injective solutions of $z + 2x = 3y$ — both patterns at once — is given in Appendix A.2, verified exhaustively against the equation; hence

$$R_4(z + 2x = 3y) \geq 57,$$

refuting the reported values $R_4(z + 2x = 3y) = 44$ and $N_{\max}(3, 4) = 43$ of [24].

Chapter 3

The Simplex Reduction and the Records

The closed form of Theorem 2.15 is a cheap certificate; the true power of the weight method lies in the grid itself. This chapter develops the reduction of the symmetric lower-bound problem to the type simplex (Section 3.1), proves that the one-weight colourings exhaust the symmetric class — so the apparent hierarchy above them is flat (Section 3.2) — and harvests the records $\text{HJ}(3, 3) \geq 22$ and $\text{HJ}(4, 2) \geq 14$, each by a single weight (Section 3.3). Section 3.4 defines the symmetric threshold, the block-symmetric ladder above it, and the extremality conjecture the records support. Throughout, symmetric colourings, their descents $\bar{c}$, and one-weight colourings are as in Lemma 2.4 and Definition 2.6.

3.1 The Simplex Reduction

For $1 \leq k \leq n$ and $v \in T_{n-k}^{(t)}$ the corner t -tuple

$$C_{k,v} = \{v + ke_1, \dots, v + ke_t\} \subseteq T_n^{(t)}$$

has t distinct elements. Monochromatic lines are detected entirely on these tuples.

Lemma 3.1 (Symmetric reduction). *Let $c = \bar{c} \circ \text{type}$ and $1 \leq K \leq n$. By the type identity (2.1), a line L_τ with $k = |I(\tau)|$ active coordinates and inactive type v is monochromatic under c iff $C_{k,v}$ is monochromatic under $\bar{c}$; and the following are equivalent: (i) no line of $L^{[K]}([t]^n)$ is monochromatic under c ; (ii) no corner tuple $C_{k,v}$ with $1 \leq k \leq K$ is monochromatic under $\bar{c}$. At $K = n$: c is line-free iff no corner tuple is monochromatic.*

3.2. One-Weight Colourings Are the Symmetric Colourings

Proof. Apply $\bar{c}$ to (2.1); every line of $L^{[K]}$ realises some (k, v) with $k \leq K$, and every such pair is realised (Lemma 2.5). $\square$

Remark 3.2 (Compression and certification). The number of corner tuples with $1 \leq k \leq K$ is $\sum_{k=1}^K \binom{n-k+t-1}{t-1}$, equal to $\binom{n+t-1}{t}$ at $K = n$. Thus the $\approx 1.3 \times 10^8$ points and $\approx 1.7 \times 10^{10}$ lines of $[3]^{17}$ become 171 cells and 969 tuples, and the 16,777,216 points of $[4]^{12}$ become 455 cells and 1365 tuples. A witness is published as an explicit table on the cells; its proof is Lemma 3.1 together with a finite check of the corner tuples, independent of any solver (Appendix A.1). Witnesses below were cross-checked by two solvers and, in small dimensions, lifted to the grid and tested against every line. For $t = 3$ the correspondence is the Fujimura-set device of the Polymath project [25]; the contribution here is its formulation as an equivalence *for colourings*, for every t and with the truncation K , its use as a SAT engine with verifiable certificates, and the witnesses of Section 3.3.

Lemma 3.3 (Monotonicity). *Fix $K \geq 1$. If $T_{n+1}^{(t)}$ admits an r -colouring with no monochromatic $C_{k,v}$, $1 \leq k \leq \min(K, n+1)$, then so does $T_n^{(t)}$. In particular the line-free symmetric dimensions form an initial segment.*

Proof. Put $\bar{c}'(v) = \bar{c}(v + e_1)$ on $T_n^{(t)}$; then $C_{k,v} + e_1 = C_{k,v+e_1}$ is a tuple of $T_{n+1}^{(t)}$ of the same k , left non-monochromatic by $\bar{c}$. $\square$

Symmetrising over the alphabet as well is impossible.

Proposition 3.4 (No fully symmetric line-free colourings). *For $K \geq t$, every colouring of $[t]^n$ invariant under all permutations of both the coordinates and the alphabet has a monochromatic line of $L^{[K]}$.*

Proof. Choose $k \equiv n \pmod{t}$ with $1 \leq k \leq t \leq K$ and set $a = (n - k)/t$: the corner tuple $C_{k,(a,\dots,a)}$ is a single orbit of the alphabet action S_t , hence monochromatic, and Lemma 3.1 lifts it to a monochromatic line. $\square$

3.2 One-Weight Colourings Are the Symmetric Colourings

One might expect one-weight colourings to be a strict, simpler subclass of the symmetric ones. They are not: allowing the weight to grow with n recovers *every* symmetric colouring.

3.2. One-Weight Colourings Are the Symmetric Colourings

Theorem 3.5 (One-weight = symmetric; the hierarchy is flat). *Fix $t \geq 2$, $r \geq 2$, $n \geq 1$. The colourings of $[t]^n$ of the form $c_{\omega, \chi}$ are exactly the symmetric colourings. For the radix weight*

$$\omega^* = (0, 1, p, p^2, \dots, p^{t-2}) \quad \text{with an arbitrary integer } p > n$$

(primality plays no role), the functional $\langle \omega^*, \cdot \rangle$ is injective on $T_n^{(t)}$ and carries each corner tuple $C_{k,v}$ bijectively onto a carry-free lattice corner; consequently every symmetric colouring $\bar{c}$ is realised by $\chi(\langle \omega^*, v \rangle) = \bar{c}(v)$, and $c_{\omega^*, \chi}$ is line-free iff no corner tuple is monochromatic under $\bar{c}$ (Lemma 3.1). Hence

$$n_{\text{ow}}(t, r) = n_{\text{sym}}(t, r),$$

and any colouring built from finitely many weights, $w \mapsto \Psi(\langle \omega^{(1)}, \text{type}(w) \rangle, \dots, \langle \omega^{(s)}, \text{type}(w) \rangle)$, realises only the symmetric class: there is no multi-weight tier.

Proof. On $T_n^{(t)}$ the entries satisfy $0 \leq v_a \leq n < p$ for $a \geq 2$, so $\langle \omega^*, v \rangle = \sum_{a \geq 2} v_a p^{a-2}$ is the base- p integer with digits $(v_2, \dots, v_t)$ — injective, since $v_1 = n - \sum_{a \geq 2} v_a$ is determined. A corner point $v + ke_a$ with $a \geq 2$ has a -digit $v_a + k \leq n < p$ (as $v_a \leq n - k$), so no carry occurs: $\langle \omega^*, \cdot \rangle$ maps $C_{k,v}$ bijectively onto the digit corner $\{(v_2, \dots, v_t) + k\hat{e}_a\}_{a \in [t]}$, where $\hat{e}_a$ is the a -th standard basis vector of the digit space for $a \geq 2$ and $\hat{e}_1 := 0$. Thus $\chi(\langle \omega^*, v \rangle) := \bar{c}(v)$ is well defined and $c_{\omega^*, \chi} = \bar{c} \circ \text{type}$; line-freeness is Lemma 3.1. Every $c_{\omega, \chi}$ is symmetric, so $n_{\text{ow}} \leq n_{\text{sym}}$; choosing $p > n_{\text{sym}}$ realises an extremal symmetric witness as a one-weight colouring, giving $n_{\text{ow}} \geq n_{\text{sym}}$. A multi-weight colouring factors through type, hence is symmetric, and is in turn one-weight by the above. $\square$

Remark 3.6 (Finite reach versus all dimensions). The radix weight depends on n through $p > n$, and the dependence is not an artifact of the proof: $\langle \omega, \cdot \rangle$ maps $T_n^{(t)}$ into an interval of $D_\omega n + 1$ integers, so injectivity forces $D_\omega \geq ((\binom{n+t-1}{t-1} - 1)/n = \Theta(n^{t-2})$ for $t \geq 3$; realising the full symmetric class in dimension n requires a weight of diameter polynomial in n . For a *fixed* weight and palette required to avoid lines in *every* dimension — the bracket regime of Chapter 4 — this device is unavailable, and there the one-weight class is a genuine restriction. The collapse $n_{\text{ow}} = n_{\text{sym}}$ is a statement about finite reach only. It also locates the gap between the closed form of Theorem 2.15 and the reach n_ω : where the two diverge, the difference is precisely the carry homothets — homothets whose base the grid never realises — which the interval count charges but the simplex does not.

3.3 The Records

3.3.1 $\text{HJ}(3, 3) \geq 22$ by a single weight

The previous record was $\text{HJ}(3, 3) \geq 14$ [6]. By Theorem 3.5 a single weight reaches the full symmetric value, and $29 > 21$ makes $(0, 1, 29)$ the clean choice.

Theorem 3.7 (One-weight; computer-assisted). *The weight $\omega = (0, 1, 29)$ admits a palette $\chi: \mathbb{Z} \rightarrow \{0, 1, 2\}$ for which $c_{\omega, \chi}$ has no monochromatic line on $[3]^{21}$; hence $\text{HJ}(3, 3) \geq 22$.*

Proof. Since $29 > 21$, Theorem 3.5 makes $\langle \omega, \cdot \rangle: (v_1, v_2, v_3) \mapsto v_2 + 29v_3$ injective on $T_{21}^{(3)}$, carrying each corner triple $C_{k,v}$ onto the carry-free planar corner $\{(v_2, v_3), (v_2+k, v_3), (v_2, v_3+k)\}$. By Lemma 2.14, $c_{\omega, \chi}$ is line-free iff $\bar{c}(v) := \chi(v_2 + 29v_3)$ leaves all such corners non-monochromatic. A 3-colouring of the 253 cells of $T_{21}^{(3)}$ avoiding all 1771 corner triples ($1 \leq k \leq 21$) exists (Table A.1); the dependency-free verifier of the repository reports zero monochromatic triples, cross-checked independently. The induced palette χ is supported on the 253 realised levels $\{v_2 + 29v_3 : v_2 + v_3 \leq 21\} \subset \{0, \dots, 609\}$. $\square$

Proposition 3.8 (A compact periodic witness; computer-verified). *For the weight $\omega = (0, 1, 4)$ and the 49-periodic palette*

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(listing $\psi(0), \dots, \psi(48)$), the colouring $c(w) = \psi(\langle \omega, \text{type}(w) \rangle \bmod 49)$ has no monochromatic line on $[3]^{14}$, verified by direct enumeration of all corner triples. Hence $\text{HJ}(3, 3) \geq 15$ from a 49-cell certificate, against the 120 simplex cells of $T_{14}^{(3)}$.

Remark 3.9 (The ladder at $(3, 3)$). The witnesses at $(3, 3)$ form a ladder, cheapest to strongest: the arithmetic closed form, 13; the triples $\{0, 1, 3\}$ and $\{0, 1, 4\}$, 14 (Table 2.1); the 49-cell periodic palette, 15; the length-76 interval certificate, 16 (Theorem 2.22); the 253-cell radix table, 22. Each step measures what a layer of structure buys. A (non-exhaustive) search over weights $(0, s, s+u)$ with $s \leq 6$, $s+u \leq 12$ and moduli $m \leq 64$ found no periodic palette line-free beyond $n = 14$, so the gap from 15 to 22 is genuinely the price of periodicity; Chapter 5 explains why no comparably compressed witness exists here. The record instance (253 cells, 1771 triples) sits at the satisfiability phase transition: CryptoMiniSat settles it in seconds while stalling at $n = 22$, where no solver has returned UNSAT, so the bound may still improve.

3.3.2 $\text{HJ}(4, 2) \geq 14$ in closed form

Theorem 3.10 (One-weight closed form). *Let $\omega = (0, 2, 3, 5)$ and let $\chi: \mathbb{Z}_{26} \rightarrow \{0, 1\}$ be*

$$\chi = (1, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0)$$

(listing $\chi(0), \dots, \chi(25)$). Then $c(w) = \chi(\langle \omega, \text{type}(w) \rangle \bmod 26)$ has no monochromatic line on $[4]^{13}$; hence $\text{HJ}(4, 2) \geq 14$. The reach is sharp: line-free at $n = 13$, not at $n = 14$.

Proof. The colouring is symmetric. By Lemma 3.1 ($K = n = 13$) line-freeness is no corner quadruple $C_{k,v}$ being monochromatic; the four cells carry $\chi(b), \chi(b + 2k), \chi(b + 3k), \chi(b + 5k)$ with $b = \langle \omega, v \rangle \bmod 26$. Over all (k, v) the residue tuples take 253 values, none monochromatic (verified by direct enumeration, Appendix A.1); at $n = 14$ the enlarged (k, v) -range produces a monochromatic quadruple. $\square$

Remark 3.11 (Three readings of 14). The value 14 arises three ways: as the period-26 witness above; as the closed form $\lceil (G_2(\{0, 2, 3, 5\}) - 1)/5 \rceil = 14$ with $G_2(\{0, 2, 3, 5\}) = 67$ (Table 2.1); and, since one weight equals symmetric (Theorem 3.5), as the lower bound $n_{\text{sym}}(4, 2) \geq 13$, whose frontier $n = 14$ is open. The sharpness in Theorem 3.10 is a property of the specific period-26 palette, not of the one-weight *family*, and there is no separate multi-weight tier to invoke; pushing past the symmetric reach requires the block-symmetric relaxation of Section 3.4. The weight $(0, 2, 3, 5)$ is a single $\text{AGL}(1, 13)$ -orbit and reflection-symmetric, $S = 5 - S$ — as is the second champion $\{0, 1, 5, 6\}$ (Remark 7.24); where exactly the period-26 palette exhausts its power is made precise in Chapter 5. Against the classical picture: Corollary 2.16 — tight in every exactly-known case — is *not* tight at $(4, 2)$, since it gives only 12 ($W(4, 2) = 35$): the Hales–Jewett numbers exceed their van der Waerden shadows. Both record witnesses are symmetric and neither is of sum type, a sum-type colouring being line-free only up to 12 at $(3, 3)$ and 11 at $(4, 2)$.

Lemma 3.12 (Even affine orbits are palindromic). *Let $S \subseteq \mathbb{Z}_m$ be an orbit of size $2j$ of an affine map $\alpha(x) = ux + v$ on $\mathbb{Z}_m$ with $u^j \equiv -1 \pmod{m}$. Then α^j is a reflection $x \mapsto -x + c$ preserving S , and some choice of representatives with $\min S = 0$ satisfies $S = D - S$, $D = \max S$: the cut must be made at a reflection-invariant gap of S , which always exists.*

Proof. $\alpha^j(x) = u^j x + v(u^{j-1} + \dots + u + 1) = -x + c$, an involution preserving the orbit. No element of S is fixed by it — a fixed element would have nontrivial

stabiliser in $\langle \alpha \rangle$, against $|S| = 2j$ — so each of the at most two axis positions of the reflection on the m -cycle lies inside a gap of S , and such a gap is invariant. Reading the representatives $s_1 < \dots < s_{2j}$ off an invariant gap, the reflection pairs s_i with s_{2j+1-i} ; the sums $s_i + s_{2j+1-i}$ are all congruent to c modulo m and consecutive ones differ by less than m , hence all are equal, and translating s_1 to 0 reads $S = D - S$. The proviso is genuine: the orbit of $x \mapsto 5x + 9$ on $\mathbb{Z}_{13}$ ($u = 5$, $5^2 \equiv -1$) has minimum-zero lift $\{0, 2, 6, 9\} \neq 9 - \{0, 2, 6, 9\}$, while cutting after 0 gives $\{0, 4, 7, 11\} = 11 - \{0, 4, 7, 11\}$. $\square$

Remark 3.13 (Orbit moduli). Both champions are free orbits of order-4 affine maps on $\mathbb{Z}_{13}$ ($5^2 \equiv -1$; Remark 7.24), and their standard lifts are palindromic, as Lemma 3.12 predicts. Orbit structure alone, however, distinguishes nothing: every triple $\{0, a, b\}$ with $\gcd(a, b) = 1$ is the orbit of an order-3 affine map exactly modulo $a^2 - ab + b^2$ — $x \mapsto 2x + 1$ on $\mathbb{Z}_7$ for $\{0, 1, 3\}$, $x \mapsto 3x + 1$ on $\mathbb{Z}_{13}$ for $\{0, 1, 4\}$, $x \mapsto 11x + 2$ on $\mathbb{Z}_{19}$ for $\{0, 2, 5\}$ — so the invariant separating the ratio champion $\{0, 2, 5\}$ of Theorem 2.22 from the orbit triples of ratio 14 is not orbit-hood but the size of the orbit modulus, 19 against 7 and 13. Whether small orbit moduli govern compressibility — the existence of small-modulus line-free palettes, as at $(4, 2)$ — is open.

3.4 The Symmetric Threshold and the Block Ladder

By Lemma 3.3 the symmetric Hales–Jewett number $\text{HJ}_{\text{sym}}(t, r) = n_{\text{sym}}(t, r) + 1$ is well defined and $\leq \text{HJ}(t, r)$, and deciding $\text{HJ}_{\text{sym}}(t, r) > n$ is a constraint-satisfaction problem of size $O(n^t)$. By Theorem 3.5 it is also exactly the one-weight reach.

Proposition 3.14. $\text{HJ}_{\text{sym}}(2, r) = r$ and $\text{HJ}_{\text{sym}}(3, 2) = 4$; thus $\text{HJ}_{\text{sym}}(t, r) = \text{HJ}(t, r)$ wherever the latter is known exactly.

Proof. Upper bounds are $\text{HJ}_{\text{sym}} \leq \text{HJ}$. For $(2, r)$, $c(w) = (\sum_i w_i) \bmod r$ on $[2]^{r-1}$ is symmetric and line-free; for $(3, 2)$, $c(w) = \lfloor ((\sum_i w_i) \bmod 4)/2 \rfloor$ on $[3]^3$ is symmetric and line-free. $\square$

Theorems 3.7 and 3.10 give $\text{HJ}_{\text{sym}}(3, 3) \geq 22$ and $\text{HJ}_{\text{sym}}(4, 2) \geq 14$, the classical lower bounds themselves. This agreement is the evidence for the following conjecture, the colouring analogue of the hyper-optimistic conjecture [26].

Conjecture 3.15 (Symmetric colourings are extremal). $\text{HJ}_{\text{sym}}(t, r) = \text{HJ}(t, r)$ for all $t, r \geq 2$.

Remark 3.16 (The optimism of Conjecture 3.15). Symmetric line-free colourings are scarce. Of the 1644 line-free 2-colourings of $[3]^3$ — already the critical dimension $\text{HJ}(3, 2) - 1$ — only 36 are symmetric (Table A.5); the other 1608 have at most one transposition, a 3-cycle, or no coordinate symmetry. The very sparsity that makes symmetric colourings computationally tractable is what makes the conjecture a strong bet: it asks a line-free symmetric colouring to survive in this thin family at every dimension up to $\text{HJ} - 1$. The same hope has already failed in the density setting — the Polymath *hyper-optimistic conjecture* [26, 25], that the densest line-free subset of $[k]^n$ in equal-slices measure is symmetric, is false in higher dimensions. That collapse is in the density regime, not the colouring one, so it is a caution rather than a counterexample; but it is why the conjecture is optimistic rather than routine.

To probe the conjecture, interpolate between the simplex and the grid by symmetrising *less*. Fix a partition $[n] = B_1 \cup \dots \cup B_b$; a colouring invariant under $S_{B_1} \times \dots \times S_{B_b}$ is *block-symmetric*, factoring through the block type map into $\prod_i T_{n_i}^{(t)}$, with $\prod_i \binom{n_i+t-1}{t-1}$ cells.

Lemma 3.17 (Block-symmetric reduction). *With block corner tuples $C_{k,v} = \{(v_1 + k_1 e_a, \dots, v_b + k_b e_a) : a \in [t]\}$ ($\sum_i k_i \geq 1$), a block-symmetric lift $\bar{c} \circ \text{type}_B$ is line-free iff no block corner tuple is monochromatic (with $\sum_i k_i \leq K$ for $L^{[K]}$).*

Writing $\text{HJ}_{b\text{-sym}}$ for the threshold over partitions into $\leq b$ blocks, splitting a block enlarges the class, so

$$\text{HJ}_{\text{sym}} = \text{HJ}_{1\text{-sym}} \leq \text{HJ}_{2\text{-sym}} \leq \dots \leq \text{HJ}(t, r),$$

constant iff Conjecture 3.15 holds; a strict step at any rung refutes it while improving the bound. By Theorem 3.5 the multi-*weight* classes all sit at the bottom rung, so this ladder — not weight count — is the genuine hierarchy above the symmetric reach. The first rungs are cheap: at $(3, 3)$, $n = 22$, the $(21, 1)$ partition has $3 \cdot 253 = 759$ cells and 7337 block tuples; at $(4, 2)$, $n = 14$, blocks $(7, 7)$ give 14400 cells. Both frontiers ($\text{HJ}(3, 3) \geq 23$, $\text{HJ}(4, 2) \geq 15$) await an UNSAT/SAT at these instances.

Chapter 4

Bracket and Interval Variants, and Two Conjectures

The Hales–Jewett theorem asserts the existence of a monochromatic line, saying nothing of its shape or of the line-free colourings in dimension $\text{HJ} - 1$. Restricting the line family probes both. This chapter treats the two restrictions along the active set: bounding its *size* (the bracket numbers, Sections 4.1–4.3) and bounding its *shape* (the interval numbers, Section 4.4), which turn out to be transverse axes — the first is exactly where a fixed periodic weight lives, the second provably invisible to the whole symmetric class (Proposition 4.15). Section 4.5 states the Collapse and Diagonal-only conjectures and collects the state of knowledge in three tables.

4.1 Bracket Numbers and Ceilings

Combining the line restriction $L^{[k]}$ with the class restriction of Definition 2.8, the *bracket Hales–Jewett number* $\text{HJ}_C^{[k]}(t, r)$ is the least n forcing a monochromatic line of $L^{[k]}([t]^n)$ under every C -colouring, and $= \infty$ if no such n exists. Since $L^{[K]} \subseteq L^{[K']}$ for $K \leq K'$ and $L^{[\text{HJ}]} = L$ in dimension HJ , the *bracket ceilings* κ_C are finite and inherit the class order:

$$\kappa_{\text{sum}} \leq \kappa_{\text{ow}} \leq \kappa_{\text{sym}} \leq \kappa_{\text{all}} \leq \text{HJ}(t, r) - 1.$$

A single number κ_C records the whole row $(\text{HJ}_C^{[k]})_{k \geq 1}$: it is ∞ for $k \leq \kappa_C$ and finite for $k \geq \kappa_{\text{all}} + 1$, with the intermediate range the content of Section 4.5.

Proposition 4.1. $\text{HJ}^{[1]}(t, r) = \infty$ for all $t \geq 2, r \geq 2$; indeed $\kappa_{\text{sum}}(t, r) \geq 1$.

Proof. Color w by $\sigma(w) \bmod 2$, a sum-type colouring. A line in $L^{[1]}$ has one active coordinate, so its t words have consecutive weights and alternate in colour. $\square$

4.2 The Sum-Type Ceiling

Lemma 4.2 (One-dimensional reduction). *Let $c = \chi \circ \sigma$ be sum-type. A line with k active coordinates and inactive weight S has word-weights $S+k, S+2k, \dots, S+tk$, a t -term progression of gap k , and is monochromatic under c iff χ is constant on it; the progressions realised in $[t]^n$ with $k \leq K$ are exactly the t -term progressions of gap $\leq K$ inside $[n, tn]$. Consequently, for every $K \geq 1$,*

$$\text{HJ}_{\text{sum}}^{[K]}(t, r) = \infty \iff \exists \chi: \mathbb{Z} \rightarrow [r] \text{ with no monochromatic } t\text{-term AP of gap } \leq K,$$

so $\kappa_{\text{sum}}(t, r)$ is the restricted-gap threshold of Brown–Graham–Landman [27], finite by van der Waerden and non-decreasing in t .

Proof. The weights of $\tau(a)$ are $S + ak$, and a one-weight colouring of the form $\chi \circ \sigma$ is constant on σ -fibres. A progression $\{A, A+k, \dots, A+(t-1)k\} \subseteq [n, tn]$ is realised iff its inactive weight $S = A - k$ lies in $[n-k, t(n-k)]$, which is exactly containment in $[n, tn]$. Right to left in the equivalence, restrict χ to the windows $[n, tn]$; left to right, colourings of arbitrarily long windows avoiding monochromatic t -term progressions of gap $\leq K$ exist, so Kőnig’s lemma yields such a colouring of $\mathbb{Z}$, and any monochromatic progression in $\mathbb{Z}$ lies in some finite window. $\square$

The upper bound is immediate: every r -colouring of $[1, W(t, r)]$ contains a monochromatic t -AP $\{a, \dots, a + (t-1)d\}$, and $(t-1)d \leq W(t, r) - 1$ caps its gap at $\lfloor (W(t, r) - 1)/(t-1) \rfloor$ — a gap no colouring of $\mathbb{Z}$ can avoid. With the block construction below this brackets the ceiling,

$$(t-1)r - 1 \leq \kappa_{\text{sum}}(t, r) \leq \left\lfloor \frac{W(t, r) - 1}{t-1} \right\rfloor - 1. \quad (4.1)$$

Theorem 4.3 (Periodic block colouring). *Set $b = t-1$ and $m = (t-1)r$, and let g be the m -periodic palette $0^b 1^b \dots (r-1)^b$, $g(x) = \lfloor (x \bmod m)/b \rfloor$. Under $c = g \circ \sigma$ a line of k active coordinates is monochromatic iff $m \mid k$; hence c has no monochromatic $L^{[m-1]}$ -line in any dimension, so $\text{HJ}^{[m-1]}(t, r) = \infty$ and $\kappa_{\text{sum}}(t, r) \geq (t-1)r - 1$.*

Proof. By Lemma 4.2 the line is monochromatic iff g is constant on a t -term AP of gap k . If $m \mid k$ the t weights agree mod m , one colour. If $m \nmid k$, put

$d = k \bmod m \in [1, m-1]$: each colour is an arc $B_i = \{ib, \dots, ib + b - 1\}$ of b consecutive residues mod m , and two residues in one arc differ circularly by $\leq b - 1$. With circular step $s = \min(d, m - d)$: if $s \geq b$ consecutive terms fall in different arcs; if $s < b$ they advance within an arc by the fixed signed step of size s (the unique representative of d in $(-b, b)$, as $m \geq 2b$), leaving it after $\leq \lfloor (b-1)/s \rfloor + 1 \leq b$ terms. Either way a monochromatic run has $\leq b = t - 1 < t$ terms. The cap is sharp at $k = m$. $\square$

Corollary 4.4. *For $t \geq 3, r \geq 2$: (i) $\text{HJ}^{[3]}(t, 2) = \infty$, so $\text{HJ}^{[3]}(3) = \text{HJ}^{[2]}(3) = \infty$ and Conjecture 4.22 holds at $(3, 2)$; (ii) $\text{HJ}^{[5]}(3, 3) = \infty$.*

Proposition 4.5. $\kappa_{\text{sum}}(3, 2) = 3$ and $\kappa_{\text{sum}}(3, 3) = 11$.

Proof. At $r = 2$ the two bounds of (4.1) meet at 3 ($W(3, 2) = 9$). At $r = 3$ the 12-periodic palette $(2, 0, 1, 2, 1, 1, 0, 1, 2, 0, 0, 2)$ carries no monochromatic 3-AP of gap ≤ 11 (finite check, Appendix A.2), so $\kappa_{\text{sum}}(3, 3) \geq 11$; and the longest 3-colouring avoiding monochromatic 3-APs of gap ≤ 12 has length $26 < 27 = W(3, 3)$, so every length-27 interval forces one and $K = 12$ is impossible. $\square$

The block reach is linear in t ; the multiplicative structure of $\mathbb{Z}/p$ does far better. Rabung's power-residue method [28], the standard source of strong van der Waerden lower bounds, colours residues by discrete logarithm.

Theorem 4.6 (Power-residue colouring, after Rabung). *Let p be prime, $r \mid p-1$, g a primitive root mod p , and $\chi(x) = \text{ind}_g(x) \bmod r$ for $x \not\equiv 0$, $\chi(0) = 0$, extended p -periodically. If χ has no monochromatic t -term AP of nonzero gap in $\mathbb{Z}/p$, then $\text{HJ}^{[p-1]}(t, r) = \infty$ and $\kappa_{\text{sum}}(t, r) \geq p - 1$.*

Proof. By Lemma 4.2 and p -periodicity, a line of $1 \leq k \leq p-1$ active coordinates is monochromatic iff χ is constant on a t -AP of gap $k \not\equiv 0 \pmod{p}$; the hypothesis forbids this. $\square$

Remark 4.7. As ind_g is additive, $x \mapsto d^{-1}x$ sends a monochromatic gap- d progression to a gap-1 one, so only spacing 1 need be checked — against the longest monochromatic run of $\chi(1) \cdots \chi(p-1)$, except that the artificial value $\chi(0)$ can bridge the runs flanking 0. Thus $p = 17$ fails for $t = 4$ despite having no three consecutive quadratic residues in $[1, 16]$: its Legendre colouring is constant on $\{15, 16, 0, 1\}$, consistent with $\kappa_{\text{sum}}(4, 2) \leq 10$.

Corollary 4.8. *If $p := \lfloor (W(t, r) - 1)/(t - 1) \rfloor$ is prime with $r \mid p - 1$ and Theorem 4.6 applies, its bound $p - 1$ meets the upper end of (4.1), so $\kappa_{\text{sum}}(t, r) =$*

$\lfloor (W(t, r) - 1)/(t - 1) \rfloor - 1$. This occurs at $(4, 2)$ ($p = 11$) and $(4, 3)$ ($p = 97$; $W(4, 3) = 293$, the certificate palette is reproduced in Appendix A.2):

$$\kappa_{\text{sum}}(4, 2) = 10, \quad \kappa_{\text{sum}}(4, 3) = 96.$$

Here p is the prime underlying the extremal van der Waerden colouring — for $(4, 3)$, Rabung’s 3-colouring realising $W(4, 3) > 292$ [28] — of which $\kappa_{\text{sum}} = p - 1$ is the bounded-gap shadow.

Remark 4.9. At the largest admissible prime the method gives $\kappa_{\text{sum}}(4, 2) \geq 10$, $\kappa_{\text{sum}}(5, 2) \geq 36$, $\kappa_{\text{sum}}(6, 2) \geq 138$ ($p = 11, 37, 139$), far above the block values $2t - 3 = 5, 7, 9$: the ceiling tracks the super-polynomial van der Waerden growth. Admissibility is governed by runs of power residues (tabulated for $p < 420000$ in [29]; cf. [30]), so admissible primes lie in a bounded range, and locating κ_{sum} in (4.1) becomes a question of analytic number theory. Where the plain construction trails ($36 < 43$ at $(5, 2)$, $138 < 225$ at $(6, 2)$), the Cyclic Zipper Method [31, 32] is the route to close the gap; its periodic adaptation is open. No admissible prime exists at $(3, 3)$, consistent with $\kappa_{\text{sum}}(3, 3) = 11 < 12$.

4.3 Periodic One-Weight Ceilings

By Remark 3.6 the genuine one-weight restriction appears in the bracket regime, where a single *periodic* palette must avoid bounded-complexity lines in all dimensions. Here non-arithmetic weights pass the van der Waerden ceiling.

Theorem 4.10 ($\text{HJ}^{[12]}(3, 3) = \infty$). For $\omega = (0, 5, 7)$ and the 13-periodic palette $\psi = (1, 0, 0, 1, 0, 1, 0, 0, 1, 2, 2, 2, 2)$ (listing $\psi(0), \dots, \psi(12)$), $c(w) = \psi(\langle \omega, \text{type}(w) \rangle \bmod 13)$ has no monochromatic line with ≤ 12 active coordinates, in every dimension.

Proof. The words carry $\psi(b), \psi(b + 5k), \psi(b + 7k)$; as 5 is a unit mod 13, b ranges over $\mathbb{Z}_{13}$, so it suffices to test the 13×12 pairs (b, k) , $1 \leq k \leq 12$, each bichromatic. At $k \equiv 0$ the indices coincide, so the cap is sharp. $\square$

Theorem 4.11 ($\text{HJ}^{[12]}(4, 2) = \infty$). For $\omega = (0, 2, 3, 5)$ and $\chi_0: \mathbb{Z}_{13} \rightarrow \{0, 1\}$ with $\chi_0^{-1}(1) = \{4, 5, 7, 9, 11, 12\}$, the colouring $c(w) = \chi_0(\langle \omega, \text{type}(w) \rangle \bmod 13)$ has no monochromatic line with ≤ 12 active coordinates, in every dimension.

Proof. As 2 is a unit mod 13, the 156 pairs (b, k) , $1 \leq k \leq 12$, exhaust the residue patterns of $(b, b + 2k, b + 3k, b + 5k)$, none monochromatic. $\square$

Proposition 4.12 (Periodic optimality at $(3, 3)$). *Among one-weight colourings $c(w) = \psi(\langle \omega, \text{type}(w) \rangle \bmod m)$ of $[3]^n$ with periodic $\psi: \mathbb{Z}_m \rightarrow \{0, 1, 2\}$, an exhaustive search over ω and all moduli m in the range of the repository gives largest cap $K = 12$, at $m = 13$, $\omega = (0, 5, 7)$.*

Remark 4.13 (Reach versus bracket). For full lines ($K = n$) the reach $n_\omega(t, r)$ is finite (Hales–Jewett), and by Theorem 3.5 its optimum over weights is $n_{\text{sym}}(t, r)$. The value ∞ above is a feature of the truncation: bounding $k \leq K$ leaves finitely many homothet shapes modulo a period, which a periodic palette avoids simultaneously. The chain at $(3, 3)$ is

$$11 = \kappa_{\text{sum}}(3, 3) \leq \kappa_{\text{ow}}(3, 3) \leq \kappa_{\text{sym}}(3, 3) \leq \text{HJ}(3, 3) - 1, \quad \kappa_{\text{ow}}(3, 3) \geq 12,$$

where Proposition 4.12 bounds only *periodic* palettes, so $\kappa_{\text{ow}}(3, 3) = 12$ is not claimed. This bracket ceiling must not be conflated with the finite reach, which is 21 (Theorem 3.7); how far a periodic palette can stretch *at a fixed dimension* is quantified by the anatomy of Chapter 5, where the two regimes differ by exactly one boundary scale at $(4, 2)$.

4.4 Interval Numbers

The *interval Hales–Jewett number* $\text{HJ}_C^{(q)}(t, r)$ is the least n at which every C -colouring of $[t]^n$ has a monochromatic q -fold line (∞ if none), with ceiling $\lambda_C(t, r)$. In dimension HJ every active set is a union of $\leq \lceil \text{HJ}/2 \rceil$ subintervals, so $\lambda_C \leq \lceil \text{HJ}/2 \rceil - 1$.

Proposition 4.14. $\lambda_C(t, r) \leq \kappa_C(t, r)$ for every class C , and $\lambda_{\text{sum}}(t, r) = 0$ for all t, r .

Proof. A set of $\leq q$ coordinates is a union of $\leq q$ subintervals, so $L^{[q]} \subseteq L^{(q)}$ and an $L^{(q)}$ -line-free colouring is $L^{[q]}$ -line-free, giving $\lambda_C \leq \kappa_C$. For the second claim, by Lemma 4.2 a sum-type colouring sees a q -fold line only through the gap k of its weight progression, and a single interval realises every $k \leq n$; so such a colouring is 1-fold-line-free iff its palette avoids monochromatic t -term progressions of *every* gap $\leq n$, which van der Waerden forbids for large n . $\square$

The contrast is exact: a bracket caps the gap at a fixed K , turning the sum-type problem into bounded-gap van der Waerden, while a single long interval is an unbounded gap. In fact not only sum-type colourings are blind here: the entire symmetric machinery of this thesis cannot see the interval axis.

Proposition 4.15 (Symmetric colourings are interval-blind). $\text{HJ}_{\text{sym}}^{(q)}(t, r) = \text{HJ}_{\text{sym}}(t, r)$ for every $q \geq 1$; in particular $\lambda_{\text{sym}} = \lambda_{\text{ow}} = \lambda_{\text{sum}} = 0$, and every colouring witnessing $\lambda(t, r) \geq 1$ is asymmetric in each dimension $n \geq \text{HJ}_{\text{sym}}(t, r)$.

Proof. For a symmetric colouring, monochromaticity of a line depends only on its invariant (k, v) (Lemma 3.1), and every pair (k, v) is realised by the root $*^k 1^{v_1} \dots t^{v_t}$, whose active set $[1, k]$ is a single interval. Hence a symmetric colouring has a monochromatic 1-fold line iff it has a monochromatic line at all, so the defining conditions of $\text{HJ}_{\text{sym}}^{(q)}$ and HJ_{sym} agree in every dimension; the classes ow and sum are contained in sym. $\square$

What is known about λ itself is confined to $t \leq 3$; assembled, it settles the ceiling completely there.

Example 4.16. $\lambda(2, r) = 0$: for $r = 3$, $\text{HJ}^{(1)}(2, 3) = 3$, the lower bound being $\geq \text{HJ}(2, 3) = 3$ and the upper because the four words 222, 122, 112, 111 of Proposition 2.2 ($r = 3$) are pairwise joined by single-interval lines, forcing a monochromatic 1-fold line in $[2]^3$.

Proposition 4.17 (Conlon–Kamčev [8]). $\text{HJ}^{(r-1)}(3, r) = \infty$ for odd r ; equivalently $\lambda(3, r) \geq r - 1$.

Proposition 4.18 (Leader–Räty ($r = 2$) [9]; Kamčev–Spiegel [10]). $\text{HJ}^{(r-1)}(3, r) < \infty$ for even r ; equivalently $\lambda(3, r) \leq r - 2$.

These two results, Shelah’s proof, and one monotonicity observation determine the ceiling completely.

Proposition 4.19 ($\lambda(3, \cdot)$ is determined). $\lambda(3, r) = r - 1$ for odd r and $\lambda(3, r) = r - 2$ for even r ; in particular $\lambda(3, 2) = 0$ and $\lambda(3, 3) = 2$.

Proof. Shelah’s proof yields, in every sufficiently high dimension, a monochromatic line whose active set is a union of at most $\text{HJ}(2, r) = r$ intervals (Remark 2.3; $\text{HJ}(2, r) = r$ by Proposition 2.2), so $\lambda(3, r) \leq r - 1$; for even r Proposition 4.18 improves this to $r - 2$. For the lower bounds, Proposition 4.17 gives $\lambda(3, r) \geq r - 1$ for odd r ; and $\lambda(3, \cdot)$ is non-decreasing in r , an $(r - 1)$ -colouring avoiding monochromatic q -fold lines being such an r -colouring, so for even $r \geq 4$, $\lambda(3, r) \geq \lambda(3, r - 1) = r - 2$. At $r = 2$ the lower bound is vacuous. $\square$

So $\lambda(3, \cdot)$ is settled and jumps with the parity of r — a feature absent from κ , whose van der Waerden origin is parity-blind — and by Proposition 4.15 the witnesses behind Proposition 4.17 live strictly off the simplex. What remains at

$t = 3$ is quantitative: the finite interval numbers themselves. Finiteness of the smallest, $\text{HJ}^{(1)}(3, 2)$, is Leader–Räty’s theorem [9], disproving the conjecture of [8] that it is infinite; the exact value is new.

Theorem 4.20 (Computer-assisted). $\text{HJ}^{(1)}(3) = 5$, whereas $\text{HJ}(3) = 4$.

Proof. A SAT instance with one variable per word and two clauses per line of $L^{(1)}([3]^4)$ is satisfiable; its witness, re-checked against all 142 such lines, gives $\text{HJ}^{(1)}(3) \geq 5$ (Appendix A.3). The same encoding in dimension 5 (all 2^{35} colourings, no symmetry imposed) is unsatisfiable, so $\text{HJ}^{(1)}(3) = 5$. Deletion minimisation extracts a minimal forcing subfamily of 262 lines on 172 words: the obstruction has no small core, unlike $\text{HJ}^{(1)}(2, 3)$, forced by a four-word clique. $\square$

Question 4.21. The ceilings at $t = 3$ are settled (Proposition 4.19); the finite interval numbers are not. Beyond $\text{HJ}^{(1)}(3) = 5$ — the remaining $(3, 2)$ values are trivially 4, every subset of $[4]$ being a union of at most two intervals — the first unknown value is $\text{HJ}^{(3)}(3, 3)$, where the only upper bounds come from Shelah’s argument. For $t \geq 4$ nothing is known beyond $\lambda(t, r) \leq \min\{\lceil \text{HJ}(t, r)/2 \rceil, \text{HJ}(t - 1, r)\} - 1$; by Proposition 4.15 any lower bound requires genuinely asymmetric constructions.

4.5 Two Conjectures and the State of Knowledge

Every line of $[t]^{\text{HJ}}$ lies in $L^{[\text{HJ}]}$, so $\text{HJ}^{[\text{HJ}(t, r)]}(t, r) = \text{HJ}(t, r)$ trivially.

Conjecture 4.22 (Collapse). $\text{HJ}^{[\text{HJ}(t, r)-1]}(t, r) = \infty$ for all $t \geq 2, r \geq 2$; equivalently $\kappa_{\text{all}}(t, r) = \text{HJ}(t, r) - 1$.

A colouring whose only monochromatic line is the diagonal is $L^{[\text{HJ}-1]}$ -line-free at $n = \text{HJ}$ — the diagonal alone is fully active — so it certifies the first inequality of Collapse. Such colourings exist at small n (Appendix A.4):

Conjecture 4.23 (Diagonal-only). For every $n \leq \text{HJ}(t, r)$ some r -colouring of $[t]^n$ has the diagonal $L_{*...*}$ as its only monochromatic line. Equivalently, $\text{HJ}(t, r) = \max\{n : [t]^n \text{ is diagonal-only colourable}\}$.

Proposition 4.24 (The range is maximal). For $n > \text{HJ}(t, r)$ no r -colouring of $[t]^n$ has the diagonal as its only monochromatic line.

Proof. Fix such a colouring c and a letter a , and colour $[t]^{n-1}$ by $c_a(w) = c(wa)$. Since $n - 1 \geq \text{HJ}(t, r)$, some line $L_{\tau'}$ is monochromatic under c_a ; then $\tau'a$ is a root of $[t]^n$ whose active set avoids coordinate n , and $L_{\tau'a}$ is a nondiagonal monochromatic line under c . $\square$

4.5. Two Conjectures and the State of Knowledge

The naive interval analogue of Collapse, by contrast, fails outright: it would force $\text{HJ}^{(1)}(2, 3) = \infty$ and $\text{HJ}^{(1)}(3) = \infty$, against Example 4.16 and Theorem 4.20.

Remark 4.25 (A refuted Collapse need not be fatal: the offset). Either way the bracket ceiling is informative. Write $\kappa_{\text{sym}}(t, r) = \text{HJ}(t, r) - \varphi(t, r)$ with $\varphi \geq 1$. The assertion $\varphi \equiv 1$ is the common strengthening of Conjectures 4.22 and 3.15: it implies Collapse through $\kappa_{\text{sym}} \leq \kappa_{\text{all}} \leq \text{HJ} - 1$, and, since at $n = \text{HJ} - 1$ the bracket $[\text{HJ} - 1]$ contains every line, $\kappa_{\text{sym}} = \text{HJ} - 1$ forces a symmetric line-free colouring of $[t]^{\text{HJ}-1}$, giving extremality; the converse is unclear. But any explicit upper bound on φ serves as well. Bounding $\text{HJ}(t, r)$ from above is a forcing statement over all t^n points of the grid — an unsatisfiability certificate out of reach. Bounding $\kappa_{\text{sym}}(t, r)$ from above is the same forcing restricted to bounded-scale corner tuples on the $O(n^{t-1})$ cells of the type simplex — within reach. A known offset transports the second into the first, $\text{HJ} \leq \kappa_{\text{sym}} + \varphi$, so the inaccessible upper bound would follow from the accessible one.

Table 4.1: Small Hales–Jewett numbers and upper bounds in terms of $\text{HJ}^{[k]}(t, r)$.

	exact value =: x	$\text{HJ}^{[4]}(t, r)$	$\text{HJ}^{[3]}(t, r)$	$\text{HJ}^{[2]}(t, r)$	$\text{HJ}^{[1]}(t, r)$
$\text{HJ}(1)$	1	1	1	1	1
$\text{HJ}(2)$	2	2	2	2	∞^a
$\text{HJ}(3)$	4^b	4	∞^c	∞^c	∞
$\text{HJ}(4)$	$14 \leq x < 10^{11d}$	∞^e	∞^c	∞^c	∞
$\text{HJ}(5)$	$45 \leq x^i$	∞^c	∞^c	∞^c	∞
$\text{HJ}(6)$	$227 \leq x^i$	∞^c	∞^c	∞^c	∞
$\text{HJ}(1, 3)$	1	1	1	1	1
$\text{HJ}(2, 3)$	3	3	3	∞^f	∞
$\text{HJ}(3, 3)$	$22 \leq x^g$	∞^h	∞^h	∞^h	∞

* the values are non-decreasing from left to right, by definition.

^a Proposition 4.1.

^b Hindman–Tressler [2].

^c Theorem 4.3 with $r = 2$: $\text{HJ}^{[K]}(t, 2) = \infty$ for all $K \leq 2t - 3$.

^d Lower bound by Theorem 3.10 (previously 12, Corollary 2.10); upper bound by Lavrov [4].

^e Theorem 4.11: $\text{HJ}^{[K]}(4, 2) = \infty$ for every $K \leq 12$ via the periodic one-weight $(0, 2, 3, 5)$.

^f Theorem 4.3 at $(t, r) = (2, 3)$: $\text{cap } m - 1 = r - 1 = 2$.

^g Theorem 3.7.

^h Theorem 4.10: $\text{HJ}^{[K]}(3, 3) = \infty$ for every $K \leq 12$.

ⁱ Corollary 2.10.

4.5. Two Conjectures and the State of Knowledge

Table 4.2: Small Hales–Jewett numbers and upper bounds in terms of $\text{HJ}^{(q)}(t, r)$.

	exact value =: x	$\text{HJ}^{(4)}(t, r)$	$\text{HJ}^{(3)}(t, r)$	$\text{HJ}^{(2)}(t, r)$	$\text{HJ}^{(1)}(t, r)$
HJ(1)	1	1	1	1	1
HJ(2)	2	2	2	2	2
HJ(3)	4	4	4	4	5 ^a
HJ(4)	$14 \leq x < 10^{11\text{b}}$	?	?	?	?
HJ(5)	$45 \leq x$	?	?	?	?
HJ(6)	$227 \leq x$	?	?	?	?
HJ(1, 3)	1	1	1	1	1
HJ(2, 3)	3	3	3	3	3 ^c
HJ(3, 3)	$22 \leq x^{\text{b}}$	$< \infty^{\text{e}}$	$< \infty^{\text{e}}$	∞^{d}	∞

* the values are non-decreasing from left to right, by definition.

^a SAT verification with explicit verified witness; Theorem 4.20, Appendix A.3.

^b Theorems 3.7 and 3.10.

^c K_4 -clique argument; Example 4.16.

^d Conlon–Kamčev [8]; Proposition 4.17.

^e Finite by Proposition 4.19 ($\lambda(3, 3) = 2$), via Shelah’s argument (Remark 2.3); exact values unknown, the first being $\text{HJ}^{(3)}(3, 3)$.

4.5. Two Conjectures and the State of Knowledge

Table 4.3: Bracket ceilings $\kappa_C(t, r)$ along the colouring hierarchy. Each entry says $\text{HJ}_C^{[k]} = \infty$ for all $k \leq \kappa_C$, capturing the columns $k > 4$ of Table 4.1.

(t, r)	$\text{HJ}(t, r)$	κ_{sum}	κ_{ow}	κ_{sym}	κ_{all}
(2, 2)	2	1	1	1	1
(2, 3)	3	2	2	2	2
(3, 2)	4	3	3	3	3 ^a
(3, 3)	≥ 22	11 ^b	$\geq 12^c$	$\geq 12^d$	$\geq 12^d$
(4, 2)	≥ 14	10 ^e	$\geq 12^f$	$\geq 12^d$	$\geq 12^d$
(4, 3)	$\geq 98^g$	96 ^e	$\geq 96^d$	$\geq 96^d$	$\geq 96^d$
(5, 2)	≥ 45	$\geq 36^e$	$\geq 36^d$	$\geq 36^d$	$\geq 36^d$
(6, 2)	≥ 227	$\geq 138^e$	$\geq 138^d$	$\geq 138^d$	$\geq 138^d$

* $\kappa_{\text{sum}} \leq \kappa_{\text{ow}} \leq \kappa_{\text{sym}} \leq \kappa_{\text{all}} \leq \text{HJ} - 1$, with $\kappa_{\text{sum}}(2, r) = r - 1$.

^a All four coincide at $\text{HJ} - 1 = 3$: $\kappa_{\text{sum}} = 3$ already (Theorem 4.3), so Collapse holds (Corollary 4.4).

^b Proposition 4.5.

^c Theorem 4.10, weight (0, 5, 7); optimal among periodic one-weight palettes at (3, 3) (Proposition 4.12).

^d Unconditional bound inherited from the column to its left: no construction is known to separate κ_{ow} , κ_{sym} , κ_{all} . Collapse predicts all three equal $\text{HJ} - 1$, open for every (t, r) with $\text{HJ} > 2r$.

^e Theorem 4.6 (Rabung) at $p = 11, 37, 97, 139$; at (4, 2) and (4, 3) this meets the bracket, giving $\kappa_{\text{sum}}(4, 2) = 10$ and $\kappa_{\text{sum}}(4, 3) = 96$ (Corollary 4.8).

^f Theorem 4.11, weight (0, 2, 3, 5).

^g Corollary 2.10 with $W(4, 3) = 293$.

Chapter 5

The Boundary of the $(4, 2)$ Record

The periodic one-weight champion $\omega = (0, 2, 3, 5)$ is line-free on $[4]^{13}$ (Theorem 3.10) but no palette extends it to $[4]^{14}$: the level instance $I_{\omega,14}$ below is unsatisfiable. By the radix theorem the one-weight family is exactly the symmetric class, so the $(4, 2)$ frontier proper is the symmetric simplex instance at $n = 14$ — open, and satisfiable iff some injective radix weight beats the record — while $I_{\omega,14}$ is a strictly more constrained projection of it, the wall met by this one construction. This chapter examines that wall from both sides: an *anatomy* of the record palette locating the exact source of its power (Section 5.2), and four irreducibility measures of the infeasible instance (Section 5.3), synthesised together with a rainbow companion in Section 5.4. Statements marked *computer-assisted* were verified two independent ways.

5.1 The Level Instance

By the line–pattern correspondence (Lemma 2.14) a one-weight colouring $c_{\omega,\chi}$ is line-free on $[t]^n$ iff χ leaves no realised homothet $\langle \omega, v \rangle + kS_\omega$ monochromatic. Negating feasibility one dimension past the record gives the instance.

Definition 5.1. For $\omega_1 = 0 < \dots < \omega_t$ and $M = \omega_t n$, the *level instance* $I_{\omega,n}$ over r colours has one variable per *level* $\ell \in \{0, \dots, M\}$ used and one not-monochromatic constraint per distinct homothet $\langle \omega, v \rangle + kS_\omega$ ($1 \leq k \leq n$, $v \in T_{n-k}^{(t)}$, in range). It is satisfiable iff some ω -weight colouring is line-free on $[t]^n$.

For the non-injective $\omega = (0, 2, 3, 5)$ this is a genuine identification of the symmetric simplex instance along $\langle \omega, \cdot \rangle$, so $I_{\omega,14}$ is a *restriction* of — not

equivalent to — the symmetric instance at $n = 14$, and its unsatisfiability is the weaker statement that this one weight stops at 13.

Proposition 5.2 (Computer-assisted). *For $\omega = (0, 2, 3, 5)$, $t = 4$, $r = 2$, $n = 14$: $M = 70$, and $I_{\omega,14}$ has 443 homothets, all of size 4, on 69 used levels (levels 1 and 69 unreachable), with all scales $k = 1, \dots, 14$ occurring. It is unsatisfiable. For comparison, the arithmetic weight reaches only $n = 11$ ($W(4, 2) = 35$, Corollary 2.16), so the instance sits two dimensions above the arithmetic ceiling.*

5.2 The Anatomy of the Record Palette

Before measuring why $n = 14$ is infeasible, it is worth locating exactly why $n = 13$ is feasible. The record palette is 26-periodic, so its behaviour on a homothet $(b, b+2k, b+3k, b+5k)$ depends only on $(b \bmod 26, k \bmod 26)$, and the finite table of residue patterns can be examined scale by scale, over *all* bases — realised in the grid or not.

Proposition 5.3 (Anatomy; computer-verified). *Let $\chi: \mathbb{Z}_{26} \rightarrow \{0, 1\}$ be the record palette of Theorem 3.10 and $\omega = (0, 2, 3, 5)$. Then:*

- (i) *for every scale $1 \leq k \leq 12$ and every base $b \in \mathbb{Z}_{26}$, the pattern $(b, b+2k, b+3k, b+5k) \bmod 26$ is bichromatic under χ ;*
- (ii) *at $k = 13$ the pattern is monochromatic for 24 of the 26 residues, the two survivors being $b = 0$ — the single base realised in the grid at scale 13 (from $T_0^{(4)}$) — and the unrealised $b = 13$.*

Consequently the reach decomposes as $13 = 12 + 1$: bracket cap plus one boundary scale. Part (i) says the palette is an extremal object of the bracket regime, matching the periodic ceiling $K = 12$ of Section 4.3 at every base simultaneously; part (ii) says its thirteenth scale is carried entirely by the thinning of the realised bases at the boundary of the simplex.

Remark 5.4 (What the anatomy explains). The reach record and the bracket ceiling at $(4, 2)$ are one construction seen at two scales: the mod-13 palette of Theorem 4.11 and the mod-26 record palette exploit the same $\text{AGL}(1, 13)$ -orbit structure of the weight (Remark 7.24), and the record is precisely a bracket-extremal palette granted one extra scale by the grid's boundary. The decomposition quantifies why a 26-cell certificate suffices at $(4, 2)$ — the reach needed, 13, sits exactly one scale above the periodic ceiling 12, and one scale is what the boundary provides — and why no comparably compressed witness exists at $(3, 3)$: there the reach needed is 21, nine scales above the same ceiling 12, while the boundary thinning bought only about two scales in the searched range (Remark 3.9). The

sharpness statement of Theorem 3.10 is also refined: at $n = 14$ the enlarged base set meets the failing residues, and the MaxSAT dual below shows the palette's three monochromatic quadruples there attain the global minimum over *all* 2-colourings of the levels.

5.3 Four Measures of Irreducibility

Encode $I_{(0,2,3,5),14}$ over the 69 level variables, each homothet $\{a, b, c, d\}$ contributing the not-all-equal clauses $(\bar{x}_a \vee \cdots \vee \bar{x}_d)$ and $(x_a \vee \cdots \vee x_d)$.

Support. A deletion-filter minimal unsatisfiable subset (MUS) is not canonical here: across deletion orders the cores range over 179 to 270 of the 443 homothets. The invariant is shape, not size — every core contains a homothet of every scale $k = 2, \dots, 13$ and meets nearly all 69 levels. The contradiction has no localized support.

Width. The primal graph — levels adjacent when they co-occur in a homothet — has 69 vertices, 1386 edges, and degeneracy 32, hence treewidth ≥ 32 . With roughly 60% edge density the instance admits no narrow variable-elimination order, bounding resolution width and clause space [33].

Algebra. Over $\text{GF}(2)$, assign x_ℓ to level ℓ and to a homothet C the all-equal indicator $P_C = \prod_i x_{\ell_i} + \prod_i (1 + x_{\ell_i})$, which is 1 iff C is monochromatic.

Lemma 5.5 (Indicator degree). $P_C = 1 + \sum_{\emptyset \neq S \subseteq C} \prod_{\ell \in S} x_\ell$, multilinear of degree $t - 1$; in particular $\deg P_C = 3$ for the four-level homothets.

Proof. In $\prod_i (1 + x_{\ell_i}) = \sum_S \prod_{\ell \in S} x_\ell$ the top monomial $\prod_i x_{\ell_i}$ cancels the first product mod 2, leaving 1 and the proper non-empty products. $\square$

Theorem 5.6 (Computer-assisted; independently reproduced). $I_{(0,2,3,5),14}$ has no $\text{GF}(2)$ Nullstellensatz refutation of degree 3, 4, or 5: the 443 indicators are linearly independent with $1 \notin \text{span}\{P_C\}$; 1 is not in the span of the 29,681 generators $\{P_C\} \cup \{x_\ell P_C\}$; and not in that of the 1,070,288 generators $\{mP_C : \deg m \leq 2\}$ (row rank 977,839). Its Nullstellensatz degree is therefore at least 6.

Since $\deg P_C = 3$, a degree-6 refutation must multiply generators by monomials of degree ≥ 3 : genuine high-degree reasoning, twice the constraint degree.

Dual. Against these three lower bounds the MaxSAT dual certifies the opposite. The minimum number of monochromatic homothets over all 2-colourings of the 69 levels is *three*: some colouring leaves only three, none fewer — and the record palette itself attains this minimum at $n = 14$ (Remark 5.4). A minimum correction set thus has size 3, against a minimal core of ≥ 179 and Nullstellensatz degree ≥ 6 .

5.4 Synthesis, the Symmetric Handle, and a Rainbow Companion

The instance is globally irreducible in three proof systems — resolution support and width, and polynomial calculus — yet three monochromatic homothets from feasible. This “near-miss without a compact certificate” is why such boundaries resist certification; it also accounts for the search behaviour — stochastic local search floors at three rather than zero, and a counterexample-guided loop drives the full simplex instance to $\approx 96\%$ satisfiable and stalls, there being no small core to converge on. What the four measures do *not* see is symmetry: they are computed on the level instance, the coarsest projection, while the symmetric simplex instance at $n = 14$ retains structure that a symmetry-aware refutation — orbit-level resolution, or sum-of-squares over the orbit scaffolding — could exploit. Three concrete problems remain: settle the GF(2) Nullstellensatz degree by deciding degree 6; decide the symmetric simplex instance at $n = 14$, which would either pin $n_{\text{sym}}(4, 2) = 13$ — making $\text{HJ}(4, 2) = 14$ a candidate exact value under Conjecture 3.15 — or beat the record via a radix weight; and exhibit a symmetric, orbit-level refutation of sub-exponential size.

The same instruments, turned on the anti-Ramsey dual, expose a clean complementarity along the symmetry axis.

Definition 5.7. A line of $[t]^n$ is *rainbow* if its t points receive t distinct colours; the *anti-Hales–Jewett number* $\text{ah}(t, n)$ is the least r such that every surjective r -colouring has a rainbow line — a genuine threshold, since merging colour classes destroys no rainbow line.

By the type identity (2.1) a symmetric colouring makes a line rainbow exactly when the corner tuple $C_{k,v}$ has t distinct colours, so the corner reduction and the level encoding transfer verbatim, each not-all-equal constraint becoming not-all-distinct. What changes is which colourings are extremal.

Proposition 5.8 (Symmetry is exponentially lossy for the rainbow dual). *A symmetric rainbow-free colouring of $[t]^n$ uses at most $|T_n^{(t)}| = O(n^{t-1})$ colours.*

Since $\text{ah}(t, n) > (t - 1)^n$ [34], for every $t \geq 3$ the largest symmetric rainbow-free colouring falls short of $\text{ah}(t, n) - 1$ by an exponential factor.

Proof. A symmetric colouring is constant on each of the $|T_n^{(t)}|$ type classes; the bound $\text{ah}(t, n) > (t - 1)^n$ is exponential. $\square$

This locates the role of symmetry precisely: the one-weight = symmetric class is the *tight* model for the monochromatic problem — it attains the records and conjecturally $n_{\text{sym}}(t, r) = \text{HJ}(t, r) - 1$ — and the *exponentially lossy* model for the rainbow one. The S_n action one exploits for the records must be escaped for the rainbow bound, so the rainbow side witnesses cleanly that asymmetric colourings can be exponentially richer than symmetric ones — the encouragement that the monochromatic wall, resistant to symmetric methods, may yield to asymmetric constructions off the simplex.

Proposition 5.9 (Computer-assisted; independently verified). *There is a rainbow-free 24-colouring of $[3]^4$, so $\text{ah}(3, 4) \geq 25$, improving the lower bound 24 of [34]; with that source's upper bound $\text{ah}(3, 4) \leq 27$, the value lies in $[25, 27]$.*

The witness is an explicit 24-colouring of the 81 words, checked against all 175 lines. As at the monochromatic boundary the exact value does not close quickly — the anti-Ramsey echo of the hardness above. The native algebraic frame for the all-distinct predicate is Alon's Combinatorial Nullstellensatz [35] rather than the all-equal ideal of Section 5.3; the monochromatic weight $\leftrightarrow$ van der Waerden reduction of Corollary 2.16 likewise has a rainbow twin, weight $\leftrightarrow$ anti-van der Waerden [36].

Chapter 6

Cyclic and Geometric Hales–Jewett Numbers

Relaxing the rigid motion of a combinatorial line — all active coordinates ascending in step — enlarges the family of lines and forces a monochromatic one in smaller dimension. This chapter places three such families. The *cyclic* numbers grow only *logarithmically* in r , against the at-least-linear growth of the classical numbers, so they live on an exponentially smaller scale and their colourings cannot transfer to the classical problem; the *unit-cyclic* and *geometric* numbers interpolate. Already at $(t, r) = (3, 2)$ the four families separate, $2 < 3 = 3 < 4$.

6.1 Three Families of Lines

Identify $[t]$ with $\mathbb{Z}_t$ (the letter t being 0), so $[t]^n = \mathbb{Z}_t^n$; write $\mathbb{Z}_t^\times$ for the units and $\varphi = |\mathbb{Z}_t^\times|$. A *valid direction* is a vector $v \in (\mathbb{Z}_t^\times \cup \{0\})^n \setminus \{\vec{0}\}$, and the *cyclic line* through a in direction v is

$$L_{a,v}^{\text{cyc}} = \{a + kv : k \in \mathbb{Z}_t\}.$$

Some v_i is a unit, so v has order t and $L_{a,v}^{\text{cyc}} = a + \langle v \rangle$ is a coset of an order- t cyclic subgroup; its t points are distinct and every moving coordinate runs through all of $\mathbb{Z}_t$. Two restrictions of the direction matter: a *unit line* has $v \in \{0, 1\}^n$, and a *geometric line* has $v \in \{-1, 0, 1\}^n$ — a $\pm$ -root with each $*$ ascending ($k \mapsto k$) and each $\bar{*}$ descending ($k \mapsto t+1-k$), valid since $-1 \equiv t-1$ is a unit. Every combinatorial line is the unit line of its characteristic vector, and the ascending-only geometric lines are the combinatorial ones, so

$$\{\text{combinatorial}\} \subseteq \{\text{unit}\}, \{\text{geometric}\} \subseteq \{\text{cyclic}\}. \quad (6.1)$$

For $t \geq 3$ the unit and geometric families are incomparable: a geometric line mixing an ascending and a descending coordinate is not unit (a unit multiple landing in $\{0, 1\}^n$ forces $\alpha = 1$ on the former and $t-1 \in \{0, 1\}$ on the latter), while a wrapping unit line such as $\{(0, 1), (1, 2), (2, 0)\} \subset \mathbb{Z}_3^2$ is not geometric; for $t = 2$, where $-1 \equiv 1$, all three coincide.

The corresponding numbers $\text{HJ}^{\text{cyc}}, \text{HJ}^{\text{unit}}, \text{HJ}^{\text{geom}}(t, r)$ — the least n such that every r -colouring of $\mathbb{Z}_t^n$ has a monochromatic line of the family — are finite by (6.1) and satisfy

$$\text{HJ}^{\text{cyc}}(t, r) \leq \text{HJ}^{\text{unit}}(t, r), \text{HJ}^{\text{geom}}(t, r) \leq \text{HJ}(t, r). \quad (6.2)$$

Each family has a closed-form count which, for fixed t as $n \rightarrow \infty$, grows at its own exponential rate:

$$\begin{aligned} N_{\text{comb}} &= (t+1)^n - t^n \sim (t+1)^n, & N_{\text{geom}} &= \frac{1}{2}((t+2)^n - t^n) \sim \frac{1}{2}(t+2)^n, \\ N_{\text{unit}} &= (2^n - 1)t^{n-1} \sim \frac{1}{t}(2t)^n, & N_{\text{cyc}} &= \frac{(\varphi+1)^n - 1}{\varphi} t^{n-1} \sim \frac{1}{\varphi t}((\varphi+1)t)^n, \end{aligned}$$

with bases $t+1 < t+2 \leq 2t \leq (\varphi+1)t$, mirroring (6.1).

Counts. Combinatorial lines are the roots in $\{1, \dots, t, *\}^n$ with a star. A cyclic line determines its subgroup, generated by exactly the φ unit multiples of any valid direction (distinct, as v has a unit coordinate); each order- t subgroup has t^{n-1} cosets. There are $(\varphi+1)^n - 1$ valid directions and $2^n - 1$ unit ones, the latter in singleton multiple-classes ($\alpha v \in \{0, 1\}^n$ forces $\alpha = 1$). Geometric lines arise from $\pm$ -roots in $\{1, \dots, t, *, \bar{*}\}^n$ with a starred coordinate, two per line (τ and its reflection $\bar{\tau}(k) = \tau(t+1-k)$). $\square$

6.2 The Cyclic Numbers Are Logarithmic

Proposition 6.1 ($t = 2$). $\text{HJ}^{\text{cyc}}(2, r) = \text{HJ}^{\text{unit}}(2, r) = \text{HJ}^{\text{geom}}(2, r) = \lceil \log_2(r+1) \rceil$.

Proof. In $\mathbb{Z}_2^n$ any two distinct points differ by a nonzero $\{0, 1\}$ -vector, hence form a unit line; a colouring avoids monochromatic unit lines iff it is injective, possible iff $r \geq 2^n$. The least n with $2^n \geq r+1$ is the threshold, and the three families coincide at $t = 2$. $\square$

For an odd prime $t = p$ the cyclic lines are exactly the affine lines of $\mathbb{F}_p^n$ (every nonzero residue is a unit), so a line-free set is a cap set and the polynomial method applies.

Theorem 6.2. $\text{HJ}^{\text{cyc}}(3, r) = O(\log r)$, and $\text{HJ}^{\text{cyc}}(3, 2) = 2$.

Proof. A cap set in $\mathbb{F}_3^n$ has size $\leq A\gamma^n$ with $\gamma = 2.756 < 3$ (Croot–Lev–Pach; Ellenberg–Gijswijt [37, 38]). Some class of an r -colouring of $\mathbb{F}_3^n$ has $\geq 3^n/r$ points, which exceeds $A\gamma^n$ once $n > \log(Ar)/\log(3/\gamma)$ and then contains a line. At $n = 2$ a 2-colouring has a class of ≥ 5 points, beyond the maximal cap 4 of $\text{AG}(2, 3)$, hence a line, while $\mathbb{Z}_3^1$ (a single line) is 2-colourable; so $\text{HJ}^{\text{cyc}}(3, 2) = 2$. (Chapter 7 identifies the minimum forcing subfamily, 11 of the 12 lines, Proposition 7.9.) $\square$

The order is tight, and logarithmic for every alphabet, through a line-free box.

Proposition 6.3 (Logarithmic lower bound). *The box $B = \{0, \dots, t-2\}^n$ contains no cyclic line. Covering $\mathbb{Z}_t^n$ by $r_n = \lceil n \ln t \left(\frac{t}{t-1}\right)^n \rceil$ random translates of B and colouring each point by its first containing translate gives an r_n -colouring with line-free classes; hence $\text{HJ}^{\text{cyc}}(t, r_n) > n$ and*

$$\text{HJ}^{\text{cyc}}(t, r) \geq (1 - o(1)) \frac{\ln r}{\ln \frac{t}{t-1}} \quad (r \rightarrow \infty),$$

of order $(t-1) \ln r$; the same holds for HJ^{unit} and HJ^{geom} by (6.2).

Proof. A valid direction has a unit coordinate, which runs through all of $\mathbb{Z}_t$ and so hits $t-1 \notin \{0, \dots, t-2\}$; no cyclic line lies in B or any translate. A point lies in a random translate with probability $q = \left(\frac{t-1}{t}\right)^n$, hence misses all r_n with probability $< e^{-qr_n} \leq t^{-n}$, and a union bound over the t^n points yields a cover. Finally $\ln r_n = n \ln \frac{t}{t-1} + O(\ln n)$. $\square$

Corollary 6.4. $\text{HJ}^{\text{cyc}}(t, r) = \Theta(\log r)$ for $t \in \{2, 3\}$ and $\Omega(\log r)$ for every t ; at $t = 2$ the bound $(1 - o(1)) \log_2 r$ even matches the exact $\lceil \log_2(r+1) \rceil$.

Remark 6.5. For $p \geq 5$ a full line is a p -term progression, beyond the relation $x+y+z = \vec{0}$ that the polynomial method controls, and whether $\text{HJ}^{\text{cyc}}(p, r) = O_p(\log r)$ is open. The contrast with the classical numbers is the point: $\text{HJ}(t, r) \geq r$, since $w \mapsto \sum_i w_i \bmod r$ is line-free for all $n < r$ (along a k -active line the sum advances by $k \not\equiv 0$). Growing at least linearly, the classical numbers outstrip the logarithmic cyclic ones, so a cyclic lower-bound colouring can never transfer to a classical bound.

6.3 Unit-Cyclic and Geometric Numbers

Between the two regimes exact values are scarce and computed case by case; dimension $n = 2$ has a uniform witness.

Proposition 6.6. *For $t \geq 3$ the colouring $c(i, j) = 1 \iff (i+j) \bmod t \in \{0, 1\}$ has no monochromatic unit line on $\mathbb{Z}_t^2$; hence $\text{HJ}^{\text{unit}}(t, 2) > 2$.*

Proof. Along the unit directions $(1, 0)$, $(0, 1)$, $(1, 1)$ the quantity $i + j$ advances by 1, 1, 2, running over a coset of $\langle 1 \rangle = \mathbb{Z}_t$ (rows, columns) or of $\langle 2 \rangle$ (diagonals). The former meets both $S = \{0, 1\}$ and its complement since $t \geq 3$; for the latter, $\langle 2 \rangle = \mathbb{Z}_t$ when t is odd, while for even t the even and odd cosets meet S at 0, 1 and S^c at 2, 3 (using $t \geq 4$). $\square$

Proposition 6.7 (Computer-assisted). $\text{HJ}^{\text{unit}}(3, 2) = \text{HJ}^{\text{geom}}(3, 2) = 3$, while $\text{HJ}^{\text{unit}}(4, 2) \geq 5$ and $\text{HJ}^{\text{unit}}(5, 2) \geq 6$.

Proof. Proposition 6.6 gives $\text{HJ}^{\text{unit}}(3, 2) > 2$, and the SAT instance on the 63 unit lines of $\mathbb{Z}_3^3$ is unsatisfiable, so $= 3$. For the geometric number, a 2-colouring of $[3]^2$ splits all 8 lines (so > 2), and an exhaustive search over 2-colourings of $[3]^3$ against its 49 geometric lines finds none avoiding — the any-colouring form of “3×3×3 tic-tac-toe has no draw” — so $= 3$. The bounds at $t = 4, 5$ are satisfiable instances on $\mathbb{Z}_4^4$ (960 unit lines) and $\mathbb{Z}_5^5$ (19,375), each verified independently of the encoding. $\square$

Thus the four families separate completely at the first nontrivial point,

$$\text{HJ}^{\text{cyc}}(3, 2) = 2 < \text{HJ}^{\text{unit}}(3, 2) = \text{HJ}^{\text{geom}}(3, 2) = 3 < \text{HJ}(3, 2) = 4,$$

both inequalities of (6.2) being strict at once. Table 6.1 collects the values; the picture is a hierarchy of regimes — cyclic logarithmic in r , classical at least linear, unit and geometric (incomparable for $t \geq 3$) in between.

Table 6.1: Cyclic, unit-cyclic, and geometric Hales–Jewett numbers. In each row $\text{HJ}^{\text{cyc}} \leq \text{HJ}^{\text{unit}}, \text{HJ}^{\text{geom}} \leq \text{HJ}$.

(t, r)	HJ	HJ^{geom}	HJ^{unit}	HJ^{cyc}
$(2, r)$	r	$\lceil \log_2(r+1) \rceil$		
$(3, 2)$	4	3	3	2
$(4, 2)$	≥ 14	—	≥ 5	—
$(5, 2)$	≥ 45	—	≥ 6	—

Chapter 7

The Corner Hypergraph: Forcing and Asymptotics

An upper bound $\text{HJ}(t, r) \leq n$ says that no r -colouring of $[t]^n$ avoids a monochromatic line. This chapter recasts the two-sided problem as a statement about *structures*: which subfamilies of lines force a monochromatic line under every colouring, how small and how symmetric such a family can be, and which hypergraph makes the symmetric question natural. For $t = 2$ a clique suffices and recovers $\text{HJ}(2, r) = r$ (Section 7.1). For $t \geq 3$ no small symmetric forcing structure exists and any graph reformulation must carry a coherence constraint, so forcing families are large asymmetric SAT cores (Section 7.2). Quotienting by the coordinate action identifies the symmetric such families with the simplex cores of Chapter 3 and isolates the central object: the *corner hypergraph* $C_n^{(t)}$, a linear t -uniform hypergraph whose weak chromatic threshold *is* the symmetric Hales–Jewett number, and which for $t = 3$ is exactly the corners configuration of Ajtai and Szemerédi [39] (Section 7.3). The reformulation pays twice asymptotically (Section 7.4): a two-line Lovász–Local-Lemma argument gives $\text{HJ}_{\text{sym}}(t, r) = \Omega(r^{t-1}/t)$, and a transfer from the quasipolynomial corners theorem of Jaber, Liu, Lovett, Ostuni and Sawhney [40] pins $\text{HJ}_{\text{sym}}(3, r)$ between $e^{c \log^2 r}$ and $e^{(\log r)^{O(1)}}$ — making the extremality conjecture (Conjecture 3.15) asymptotically loaded and falsifiable.¹

¹The corrected degree ceiling of Proposition 7.27 and the results of Section 7.4 originated in an external verification report on an earlier version of this thesis; the proofs given here are self-contained and were re-verified for this text, the combinatorial counterexample (Proposition 7.26) by machine.

7.1 Forcing Structures and the Case $t = 2$

A family $\mathcal{F}$ of t -subsets of a finite set X is r -forcing if every r -colouring of X makes some member monochromatic — equivalently, the t -uniform hypergraph $(X, \mathcal{F})$ has weak chromatic number $> r$. Forcing is preserved under enlargement, so $\text{HJ}(t, r) \leq n$ iff the line family $L([t]^n)$ is r -forcing, iff some subfamily is; the aim is a small or structured one. For $t = 2$ the responsible structure is a clique.

Proposition 7.1 ($t = 2$: cliques suffice). *Let $G_2(n)$ be the collinearity graph of $[2]^n$. It is the comparability graph of the Boolean lattice; its cliques are the chains, the largest with $n + 1$ vertices; and $\chi(G_2(n)) = n + 1$, whence $\text{HJ}(2, r) = r$.*

Proof. A line in $[2]^n$ is a comparable pair $w < w'$, so adjacency is comparability and cliques are chains, the longest of length $n + 1$. A colouring is line-free iff each class is an antichain; by Mirsky's theorem [41] the fewest antichains covering the lattice is the longest chain, $n + 1$. So a line-free r -colouring exists iff $r \geq n + 1$. $\square$

Remark 7.2. At $n = r$, a maximal chain $w_0 < \dots < w_r$ and its $\binom{r+1}{2}$ lines $\{w_i, w_j\}$ force on $r + 1$ of the 2^r words — the pigeonhole certificate of Proposition 2.2. Comparability graphs being perfect, clique number equals chromatic number: the simplest forcing structure is optimal. The rest of the chapter is how this fails for $t \geq 3$.

7.2 Obstructions for $t \geq 3$

Fix $t \geq 2, n \geq 2$. The intersection graph $G([t]^n)$ has vertex set $L([t]^n)$, with $l_1 l_2$ an edge when $l_1 \cap l_2 \neq \emptyset$.

Lemma 7.3 (Linearity). *Two distinct words lie on at most one common line; hence two distinct lines meet in at most one word, and each edge of $G([t]^n)$ has a well-defined intersection word.*

Proof. From $x = \tau(a), y = \tau(b), a \neq b$, the root τ is recovered (active set where x, y differ; constants elsewhere). $\square$

A colouring c induces $\tilde{c}(e) = c(x_e)$ on the edges, x_e the intersection word; the star at a vertex l is its set of incident edges.

Lemma 7.4 (Lines through a point). *For $x \in [t]^n$, the number of lines through x is $\lambda(x) = \sum_a (2^{m_a(x)} - 1)$ with $m_a(x) = |\{i : x_i = a\}|$, and $n \leq \lambda(x) \leq 2^n - 1$; the lower bound is an equality iff the letters of x are pairwise distinct.*

Lemma 7.5 (Dictionary). *A line l is monochromatic under c iff the star at l is monochromatic under $\tilde{c}$. Hence $\text{HJ}(t, r) \leq n$ iff every induced r -edge-colouring of $G([t]^n)$ has a monochromatic star.*

Proof. Edges at l biject with pairs (word $x \in l$, line $l' \neq l$ through x), carrying $c(x)$; every $x \in l$ has $\lambda(x) \geq n \geq 2$, so every word of l is represented. $\square$

The dictionary quantifies over *induced* colourings only: edges with the same intersection word must share a colour (*coherence*). This restriction is not a convenience but the entire content, in the following quantitative sense.

Proposition 7.6 (Coherence is indispensable). *Every star of $G([t]^n)$ has at least $t(2^{\lceil n/t \rceil} - 2)$ edges. Consequently, for a uniformly random — not necessarily coherent — r -edge-colouring of $G([t]^n)$,*

$$\Pr[\exists \text{ monochromatic star}] \leq (t+1)^n r^{1-t(2^{\lceil n/t \rceil}-2)} \xrightarrow{n \rightarrow \infty} 0,$$

so for all large n (already $n \geq 12$ at $t = 3, r = 2$) non-coherent r -edge-colourings with no monochromatic star exist in abundance. Any graph-theoretic reformulation of HJ must therefore carry the coherence structure — or work with the level hypergraphs of Section 7.3, where it is built in.

Proof. In any word x some letter occurs $m_a \geq \lceil n/t \rceil$ times, so $\lambda(x) \geq 2^{\lceil n/t \rceil} - 1$ (Lemma 7.4); the star at l has $\sum_{x \in l} (\lambda(x) - 1) \geq t(2^{\lceil n/t \rceil} - 2)$ edges. A fixed star of size δ is monochromatic with probability $r^{1-\delta}$; the union bound runs over the $(t+1)^n - t^n < (t+1)^n$ lines. At $t = 3, r = 2, n = 12$: $4^{12} \cdot 2^{1-42} = 2^{-17} < 1$. $\square$

The pigeonhole certificate behind $\text{HJ}(2, r) = r$ is the clique $K_{r(t-1)+1}^{(t)}$: among $r(t-1) + 1$ words pairwise spanning lines, some t share a colour. The minimal hyperclique whose absence excludes all of these, for every r , is $K_{t+1}^{(t)}$; linearity forbids it.

Proposition 7.7 (No hypercliques). *For $t \geq 3$ the line hypergraph of $[t]^n$ contains no $K_s^{(t)}$ with $s \geq t+1$. In particular $[3]^4$ has no $K_4^{(3)}$: no pigeonhole certificate for $\text{HJ}(3, 2) \leq 4$ exists.*

Proof. Two edges of $K_s^{(t)}$ on a common $(t-1)$ -set share $t-1 \geq 2$ vertices, forbidden by Lemma 7.3; and $K_s^{(t)} \supseteq K_{t+1}^{(t)}$ for $s \geq t+1$. $\square$

The next candidate is the smallest non-2-colourable 3-uniform hypergraph, the Fano plane $PG(2, 2)$ [42]; it is linear, so the bar is not linearity but characteristic 3.

Proposition 7.8 (No Fano plane). *The Fano plane embeds into the affine lines of $\text{AG}(n, 3)$ for no n ; hence into none of the combinatorial, unit, or cyclic line hypergraphs of $[3]^n \cong \mathbb{Z}_3^n$.*

Proof. Three points of $\mathbb{F}_3^n$ are collinear iff they sum to $\vec{0}$. Summing the seven incidences through p_i gives $3p_i + \sum_{j \neq i} p_j = \vec{0}$, i.e. $p_i = S$ for all i , contradicting distinctness. $\square$

So no small symmetric configuration forces a line: the forcing families are large asymmetric cores. SAT extracts minimal ones (no proper subfamily forces) by guarding each line clause with a selector and running a deletion filter; minimality is not minimality of size, so several deletion orders are tried and each family is re-verified independently.

Proposition 7.9 (Computer-assisted; exact). *Among the 12 cyclic lines of $\mathbb{Z}_3^2$: every 2-colouring leaves ≥ 2 monochromatic, and every pair is the exact monochromatic set of some colouring. A subfamily is 2-forcing iff it omits ≤ 1 line, so the minimum has 11 lines.*

Proposition 7.10 (Computer-assisted). *The filter yields (i) a minimal 2-forcing family of 31 unit lines of $\mathbb{Z}_3^3$ on 24 words, certifying $\text{HJ}^{\text{unit}}(3, 2) \leq 3$; (ii) a minimal 2-forcing family of 67 combinatorial lines of $[3]^4$ on 52 of the 81 words, certifying $\text{HJ}(3, 2) \leq 4$. The extremal bound gives ≥ 7 lines, so geometry forces far more than the abstract minimum: forcing in grids is global.*

7.3 The Quotient and the Corner Hypergraph

Confining the search to S_n -invariant families and quotienting $G([t]^n)$ by the coordinate action merges the star dictionary (Lemma 7.5) and the symmetric reduction (Lemma 3.1) into one finite object.

Definition 7.11. The *quotient multigraph* $Q([t]^n) = G([t]^n)/S_n$ has vertices the line-orbits — equivalently the pairs (k, v) , i.e. the corner tuples $C_{k,v}$ — and edges the orbits of intersecting pairs; each edge carries the *intersection cell* $\iota(E) \in T_n^{(t)}$, and the *star* at a vertex is its incident edges. There are $\binom{n+t-1}{t}$ vertices. Q has loops: e.g. the lines $*11 \cdots 1$ and $1*1 \cdots 1$ share the invariant $(1, (n-1)e_1)$ and meet at $11 \cdots 1$.

Lemma 7.12 (Corner linearity). *Distinct corner tuples share at most one cell. Consequently all parallel edges between two distinct vertices of $Q([t]^n)$ carry the same intersection cell, and edge multiplicity records only orbit-inequivalent geometric realisations.*

Proof. Suppose $C_{k,v} \neq C_{k',v'}$ share two cells, $v + ke_a = v' + k'e_{a'}$ and $v + ke_b = v' + k'e_{b'}$ with $a \neq b$ (hence $a' \neq b'$). Subtracting, $k(e_a - e_b) = k'(e_{a'} - e_{b'})$; comparing supports and signs (as $k, k' \geq 1$) gives $a = a'$, $b = b'$, $k = k'$, hence $v = v'$, a contradiction. $\square$

A symmetric colouring induces $\tilde{c}_Q(E) = \bar{c}(\iota(E))$; for $n \geq 2$ this bijects cell colourings with induced edge-colourings, coherence surviving on a polynomial object.

Proposition 7.13 (Quotient dictionary). *For $\bar{c}$ with lift c and a line l of invariant (k, v) : the star at $[l]$ is monochromatic iff $C_{k,v}$ is monochromatic iff every line of $[l]$ is monochromatic. Hence symmetric line-free r -colourings of $[t]^n$ correspond to induced edge-colourings of $Q([t]^n)$ with no monochromatic star, and $n_{\text{sym}}(t, r) < n$ iff every such colouring has one.*

Proof. The intersection cells of the edges at $[l]$ are exactly $C_{k,v}$ (by (2.1)); then Lemma 3.1. $\square$

Corollary 7.14. *Minimal symmetrically r -forcing families are the minimal unsatisfiable subsets of the frontier simplex instances of Section 3.4 (276 cells, 2024 tuples at $(3, 3)$, $n = 22$; 680, 2380 at $(4, 2)$, $n = 14$), proving $n_{\text{sym}}(t, r) < n$, not $\text{HJ}(t, r) \leq n$.*

A symmetrically forcing family read in the grid may or may not force all colourings — a smaller question, on its word support. At the smallest instance with content this resolves completely.

Proposition 7.15 (Lifted cores). *If every r -colouring of $W(\mathcal{F}) = \bigcup_{l \in \mathcal{F}} l$ makes some $l \in \mathcal{F}$ monochromatic, then $\mathcal{F}$ is r -forcing in $[t]^n$.*

Proposition 7.16 (Computer-assisted; exact). *Let $(t, r) = (3, 2)$, $n = 4$ and $\mathcal{D} = \{C_{1,(1,1,1)}, C_{2,(0,1,1)}, C_{2,(1,0,1)}, C_{2,(1,1,0)}\}$. A family of corner triples is 2-forcing on $T_4^{(3)}$ iff it omits at most one triple, lying in $\mathcal{D}$; there are exactly four minimal symmetrically 2-forcing families, each of 19 triples, and minimal = minimum. The same boundary holds verbatim at the grid level, so each lifts to an S_4 -invariant 2-forcing family certifying $\text{HJ}(3, 2) \leq 4$; the smallest has 151 lines on the 81 words.*

Proof. Forcing being monotone, the 210 single- and double-omission patterns were decided on the 15 cells by exhaustion and on the 81 words by two solvers, agreeing throughout; exactly the four omissions from $\mathcal{D}$ remain forcing. $\square$

Remark 7.17. At $(3, 2)$, $n = 4$ the forcing boundary is identical on simplex and grid — the propagation Conjecture 3.15 needs, in stronger form. The price of invariance against Proposition 7.10(ii) is 151 lines on 81 words versus 67 on 52, a factor ≈ 2.3 . The block-symmetric ladder of Section 3.4 interpolates toward the grid; cores transport down it and across dimensions via $C_{k,v} \mapsto C_{k,v+e_1}$.

The vertex-stars of $Q([t]^n)$, read through their intersection cells, are exactly the corner tuples; forgetting the graph and keeping the tuples produces the chapter's central object.

Definition 7.18 (Corner hypergraph). $C_n^{(t)}$ is the t -uniform hypergraph with vertex set $T_n^{(t)}$ — equivalently, in the coordinates $q = (v_2, \dots, v_t)$, the lattice simplex $\{q \in \mathbb{Z}_{\geq 0}^{t-1} : |q| \leq n\}$ — and edge set the corner tuples $C_{k,v}$, i.e. the positive corners $\{q, q + k\hat{e}_1, \dots, q + k\hat{e}_{t-1}\}$, $k \geq 1$. For $t = 3$ these are the upward triangles of the Polymath–Fujimura setting (Remark 3.2) and the corners configuration of Ajtai–Szemerédi [39].

Lemma 7.19 (Pencils are complete). *For $n \geq 2$ a cell $u \in T_n^{(t)}$ lies in exactly n corner tuples, and any two are joined by an edge of intersection cell u ; its pencil is K_n .*

Proof. $u \in C_{k,v}$ iff $v = u - ke_a \geq 0$; the admissible pairs (k, a) number $\sum_a u_a = n$. Two such tuples give distinct lines through a fixed word x of type u , meeting only at x . $\square$

Lemma 7.20 (Structure). $C_n^{(t)}$ is a linear t -uniform hypergraph: every cell lies in exactly n edges (Lemma 7.19) and distinct edges share at most one cell (Lemma 7.12). It contains no $K_{t+1}^{(t)}$ (Proposition 7.7 applies verbatim through the realisation of Proposition 7.26) and, for $t = 3$ in the affine-line setting, no Fano configuration (Proposition 7.8).

For a weight ω with $|S_\omega| = t$ the same object appears one projection lower, on the integers: the level hypergraph $\mathcal{H}_\omega(n)$ has vertex set the realised levels $\Lambda_\omega(n) = \langle \omega, T_n^{(t)} \rangle$ and edge set the realised homothets $\{b + kS_\omega : 1 \leq k \leq n, b \in \langle \omega, T_{n-k}^{(t)} \rangle\}$. The set $\Lambda_\omega(n)$ is contained in an interval of $(\omega_t - \omega_1)n + 1$ integers but need not be one: for $\omega = (0, 2, 3, 5)$, $n = 14$, it has 69 elements inside $\{0, \dots, 70\}$, missing exactly $\{1, 69\}$ (Proposition 5.2) — the carry-homothet phenomenon of Remark 3.6.

Theorem 7.21 (Reformulation). *For $n \geq 2$ and any weight ω with $|S_\omega| = t$:*

- (i) $c_{\omega, \chi}$ is line-free on $[t]^n$ iff χ is a proper (weak) colouring of $\mathcal{H}_\omega(n)$; hence $n_\omega(t, r) = \max\{n : \chi_w(\mathcal{H}_\omega(n)) \leq r\}$.
- (ii) For the radix weight ω^* ($p > n$), the level map is injective on cells and $\mathcal{H}_{\omega^*}(n) \cong C_n^{(t)}$; hence, by Theorem 3.5,

$$\text{HJ}_{\text{sym}}(t, r) = \min\{n : \chi_w(C_n^{(t)}) > r\}.$$

- (iii) $\text{HJ}(t, r) \leq n$ iff every coherent (induced) r -edge-colouring of $G([t]^n)$ has a monochromatic star; the symmetric restriction is the same statement in $Q([t]^n)$.

The symmetric Hales–Jewett theorem is thus equivalent to $\chi_w(C_n^{(t)}) \rightarrow \infty$: the colouring corners theorem for $t = 3$, and the colouring Gallai–Witt theorem for the simplex pattern $\{0, \hat{e}_1, \dots, \hat{e}_{t-1}\} \subset \mathbb{Z}^{t-1}$ in general.

Proof. (i) is Lemma 2.14 at $K = n$; (ii) is Theorem 3.5, the carry-free bijection identifying corner tuples with the edges of $C_n^{(t)}$; (iii) is Lemma 7.5 and Proposition 7.13. $\square$

Proposition 7.22 ($t = 2$ is a clique, and one-weight is tight). *Let $t = 2$, $\omega_1 \neq \omega_2$. Then $\mathcal{H}_\omega(n)$ is the complete graph K_{n+1} on the levels $V_i = \langle \omega, (n-i, i) \rangle$, $0 \leq i \leq n$. Hence $c_{\omega, \chi}$ is line-free on $[2]^n$ iff $r \geq n + 1$, so $n_\omega(2, r) = r - 1 = \text{HJ}(2, r) - 1$; the obstruction is the clique K_{n+1} .*

Proof. For $0 \leq i < j \leq n$ set $k = j - i$, $v = (n - j, i)$; Lemma 2.14 gives the edge $\{V_i, V_j\}$, and $\chi(K_{n+1}) = n + 1$. $\square$

Remark 7.23 (No clean obstruction for $t \geq 3$). For $t \geq 3$ and $|S_\omega| = t$ (the hypothesis is needed: a degenerate weight such as $(0, 1, 1)$ produces two-level edges and a K_{n+1} -type problem) every edge of $\mathcal{H}_\omega(n)$ has $t \geq 3$ vertices, so Theorem 7.21(i) is a Property-B question, not a graph colouring, and the clique has no counterpart: linearity bars the complete substructure and even the Fano configuration (Lemma 7.20). Forcing is carried by homothet incidences with no bounded local certificate, and one-weight stops being tight in the *bracket* sense: the arithmetic weight reaches only van der Waerden’s ceiling, the best periodic weights $K = 12$ (Section 4.3). For finite reach, however, Theorem 3.5 restores everything — a single radix weight realises the symmetric witness $\text{HJ}(3, 3) \geq 22$ (Theorem 3.7).

Remark 7.24 (Records as affine orbits). For $(4, 2)$, scanning non-arithmetic weights with entries ≤ 6 singles out exactly $\omega = (0, 2, 3, 5)$ and $\omega = (0, 1, 5, 6)$ as reaching $n = 13$ (Theorem 3.10), others stalling at $n = 9$. Both are single

orbits of an order-4 affine map on $\mathbb{Z}_{13}$ ($5^2 \equiv -1$): $\{0, 3, 5, 2\}$ under $x \mapsto 5x + 3$ and $\{0, 1, 6, 5\}$ under $x \mapsto 5x + 1$. The level set is then one $\text{AGL}(1, 13)$ -orbit, vertex-transitive, which is what lets the period-26 palette two-colour the stars up to $n = 13$. The reflection symmetry $S = D - S$ of both champions is forced by this orbit structure (Lemma 3.12).

How close can $Q([t]^n)$ be to a *clean* forcing graph — cells as edges, corner tuples as vertex-stars, no side constraint? The answer has a global and a local half, and they point in opposite directions.

Proposition 7.25 (Global clean model exists only at $n = 2$). *A graph whose edges biject with the cells of $T_n^{(t)}$ and in which every corner tuple is a single vertex-star exists iff $n = 2$, in which case it is K_{t+1} .*

Proof. A cell lies in n tuples (Lemma 7.19) but its edge in only 2 stars, forcing $n = 2$; at $n = 2$ the $t + 1$ corner tuples on K_{t+1} realise each as a star. $\square$

Call $H \subseteq Q([t]^n)$ *cell-rainbow* if ι is injective on $E(H)$; then every edge-colouring of H is coherent. The clean $n = 2$ model does not disappear in higher dimensions — it persists *locally* in every dimension.

Proposition 7.26 (A clean K_{t+1} in every dimension; machine-verified in $Q([3]^5)$). *For every $t \geq 2$ and $n \geq 2$ let $v_0 = (n - 2)e_1 \in T_{n-2}^{(t)}$ and consider the $t + 1$ corner tuples*

$$B = C_{2,v_0}, \quad A_a = C_{1,v_0+e_a} \quad (a \in [t]).$$

Then $A_a \cap B = \{v_0 + 2e_a\}$ and $A_a \cap A_b = \{v_0 + e_a + e_b\}$ for $a \neq b$; the $\binom{t+1}{2}$ shared cells are pairwise distinct; and each shared cell is realised by an edge of $Q([t]^n)$. Consequently $Q([t]^n)$ contains a loopless, t -regular, cell-rainbow K_{t+1} for every $n \geq 2$ — the $n = 2$ model of Proposition 7.25 transported by $C_{k,v} \mapsto C_{k,v+e_1}$.

Proof. The intersection identities and the distinctness of the shared cells are immediate from $v_0 + e_a + e_b = v_0 + 2e_c \iff a = b = c$ and comparison of supports. For realisability: if a cell u lies in two tuples $C_{k,v} \neq C_{k',v'}$, fix any word x of type u ; since $u = v + ke_a$ we have $u_a \geq k$, so activating k coordinates of x carrying the letter a gives a line with invariant (k, v) through x , and likewise for (k', v') ; the two lines are distinct (distinct invariants), pass through x , hence meet exactly at x (Lemma 7.3), giving an edge of Q with intersection cell u . Choosing one such edge per pair yields the subgraph, whose vertex-stars carry exactly the tuples B and A_a . $\square$

The global counterpart is a counting ceiling — and only a global one.

Proposition 7.27 (Cell-rainbow ceiling). *Every cell-rainbow subgraph $H \subseteq Q([t]^n)$ has $|E(H)| \leq \binom{n+t-1}{t-1} = \frac{t}{n} |V(Q)|$. Hence the average degree of H over the full vertex set of Q is at most $2t/n$, and a spanning loopless d -regular cell-rainbow subgraph has $d \leq 2t/n$.*

Proof. ι is injective on $E(H)$, so $|E(H)| \leq |T_n^{(t)}|$; and $\binom{n+t-1}{t-1} / \binom{n+t-1}{t} = t/n$. $\square$

Together: no global clean graph model exists above $n = 2$ — any structure carrying all corner data must be spanning, and is then degree-starved — while the clean K_{t+1} persists locally in every dimension. Coherence-free substructures are either local or sparse; the forcing data as a whole never fits in a graph.

7.4 Asymptotic Bounds from the Corner Hypergraph

The reformulation of Theorem 7.21(ii) is not only an organising principle; it changes which tools apply. Two are recorded here: the Lovász Local Lemma, which the linearity of $C_n^{(t)}$ makes effective, and the corners theorem, which transfers verbatim.

Theorem 7.28. *For all $t \geq 2$, $r \geq 2$,*

$$\text{HJ}(t, r) \geq \text{HJ}_{\text{sym}}(t, r) \geq 2 + \left\lfloor \frac{r^{t-1}/e - 1}{t} \right\rfloor = \Omega\left(\frac{r^{t-1}}{t}\right).$$

Proof. Colour $T_n^{(t)}$ uniformly at random with r colours. The bad events are the corner tuples being monochromatic, each of probability $p = r^{1-t}$ and depending only on its t cells, hence mutually independent of all events sharing no cell. Each cell lies in exactly n tuples (Lemma 7.19) and distinct tuples share at most one cell (Lemma 7.12), so each event depends on at most $d = t(n-1)$ others. If $ep(d+1) \leq 1$, i.e. $t(n-1) + 1 \leq r^{t-1}/e$, the symmetric Local Lemma yields a proper colouring of $C_n^{(t)}$, hence (Theorem 7.21(ii)) a symmetric line-free colouring of $[t]^n$ and $\text{HJ}_{\text{sym}}(t, r) > n$; the largest such n is $1 + \lfloor (r^{t-1}/e - 1)/t \rfloor$. $\square$

Remark 7.29. Three readings. (i) This rederives the $\Omega_t(r^{t-1})$ growth of the van der Waerden transfer (the order of Corollary 2.16 with the bounds of [14, 15]) in two lines and with no arithmetic input; the quotient is precisely a dependency collapse — on the grid's line hypergraph the degree of a line is exponential in n (Proposition 7.6) and the Local Lemma yields only a logarithmic bound,

while on the simplex the degree is linear in n , boosting the bound to r^{t-1} . (ii) Moser–Tardos makes the bound constructive and supplies principled warm starts for the block-ladder pipeline of Section 3.4. (iii) The same count on a level window gives, for every t -point S of diameter D , $G_r(S) = \Omega(D r^{t-1}/t^2)$ — the asymptotic companion to the exact Gallai values of Table 2.1. The constants are asymptotic in r : at $(3, 3)$ the bound is vacuous.

Write $N_\angle(\rho)$ for the least N such that every ρ -colouring of $[0, N]^2$ contains a monochromatic corner $\{(x, y), (x + d, y), (x, y + d)\}$ with $d \neq 0$.

Lemma 7.30 (Sign reduction and transfer). *Every r -colouring of $[0, m]^2$ with $m \geq N_\angle(r^2)$ contains a monochromatic positive corner ($d = k \geq 1$); consequently*

$$\text{HJ}_{\text{sym}}(3, r) \leq 3 N_\angle(r^2).$$

Proof. Given χ , define the r^2 -colouring $\hat{\chi}(x, y) = (\chi(x, y), \chi(m - x, m - y))$. A monochromatic $\hat{\chi}$ -corner with $d > 0$ gives, through the first coordinate, a monochromatic positive χ -corner. One with $d < 0$, say $d = -k$, gives through the second coordinate the triple $(m - x, m - y), (m - x + k, m - y), (m - x, m - y + k)$ — a positive corner, inside $[0, m]^2$ since $x + d, y + d \geq 0$ — again χ -monochromatic. For the transfer, embed $[0, m]^2$ into the simplex $T_{3m}^{(3)}$ in the corner coordinates $q = (v_2, v_3)$: a positive corner $\{q, q + k\hat{e}_1, q + k\hat{e}_2\}$ with $q \in [0, m]^2$, $k \leq m$ has $|q| + k \leq 3m$, so it is an edge of $C_{3m}^{(3)}$. Every r -colouring of $T_{3m}^{(3)}$ restricts to the square; for $m \geq N_\angle(r^2)$ the restriction has a monochromatic positive corner, so $\chi_w(C_{3m}^{(3)}) > r$ and Theorem 7.21(ii) gives $\text{HJ}_{\text{sym}}(3, r) \leq 3m$. $\square$

Theorem 7.31 (Two-sided quasipolynomial bounds). *There are absolute constants $c, C > 0$ with*

$$\exp(c \log^2 r) \leq \text{HJ}_{\text{sym}}(3, r) \leq \exp((\log r)^C).$$

Proof. Upper bound. Jaber–Liu–Lovett–Ostuni–Sawhney [40] prove that corner-free subsets of $[N]^2$ have density at most $\exp(-(\log N)^{c_0})$ for an absolute $c_0 > 0$; a colour class of density $\geq 1/\rho$ is therefore forced to contain a corner once $\rho \leq \exp((\log N)^{c_0})$, i.e. $N_\angle(\rho) \leq \exp((\log \rho)^{1/c_0})$. Apply Lemma 7.30 with $\rho = r^2$.

Lower bound. By Behrend’s construction [43] there is a 3-AP-free $B \subseteq [N]$ with $|B| \geq N e^{-c_1 \sqrt{\log N}}$. Choosing $m = O((N/|B|) \log N) = e^{O(\sqrt{\log N})}$ uniform random translates of B covers $[N]$ with positive probability; colouring each integer by its first covering translate partitions $[N]$ into m classes, each a subset of a translate of B , hence 3-AP-free. Colouring the levels of the arithmetic weight

$(0, 1, 2)$ on $[0, 2n]$ by this partition gives a sum-type — in particular symmetric — line-free colouring of $[3]^n$ with $r = e^{O(\sqrt{\log n})}$ colours (Corollary 2.16: lines map to 3-term progressions); inverting, $\text{HJ}_{\text{sym}}(3, r) \geq e^{c \log^2 r}$. This is the Exactly- N protocol of Chandra–Furst–Lipton [44]. $\square$

Corollary 7.32 (The extremality conjecture is asymptotically loaded). *If $\text{HJ}_{\text{sym}} = \text{HJ}$ (Conjecture 3.15), then $\text{HJ}(3, r) \leq \exp((\log r)^{O(1)})$ — an enormous improvement over the best known upper bound $2^{2^{O(r)}}$ [5]. Conversely, any future lower bound $\text{HJ}(3, r) \geq \exp((\log r)^{\omega(1)})$ refutes the conjecture for large r . The conjecture is therefore falsifiable on the asymptotic axis, not only at the frontier instances of Section 3.4.*

Remark 7.33 (Removal, density, and the boundary). Three further transfers. (i) For $t = 4$, $C_n^{(4)}$ is the three-dimensional corners configuration, for whose colouring version [40] gives the first nontrivial bounds; through Theorem 7.21(ii) these become the first nontrivial upper bounds on $\text{HJ}_{\text{sym}}(4, r)$ and place the $(4, 2)$, $n = 14$ frontier inside the 4-player number-on-forehead communication line. (ii) The corners theorem for $t = 3$ follows from triangle removal [45], and for general t the simplex-corners density theorem follows from hypergraph removal [46]; hence $\chi_w(C_n^{(t)}) \rightarrow \infty$ has a self-contained density proof — a proof of the *symmetric* Hales–Jewett theorem that never invokes Hales–Jewett. (iii) The removal phenomenon also names what the boundary chapter observes: a colouring of $T_n^{(3)}$ with few monochromatic tuples can be corrected on few cells, and the MaxSAT distance 3 of Chapter 5 is this stability in miniature; an effective removal lemma for $C_n^{(t)}$ is the natural formalisation of the observed near-satisfiability.

Remark 7.34 (The infinite grid, and the open question). The dimension is external throughout; on the inverse limit $[t]^{\mathbb{N}} = \varprojlim_n [t]^n$, a Cantor space, continuous r -colourings factor through finite windows, and $\text{HJ}(t, r) \leq n$ is equivalent to every continuous colouring having a monochromatic limit line with active set inside a fixed n -coordinate window — the window being essential, since a colouring factoring through π_m makes every line disjoint from $[m]$ trivially monochromatic. The limit graph is again linear, not locally finite, and coherence survives in every window (Proposition 7.6): the limit reorganises rather than removes the difficulty. What it isolates is the right open question.

Question 7.35. Is there an intrinsic parameter of $G([t]^n)$ (or of its infinite-dimensional limit with the window filtration), not referring to the words, whose growth is equivalent to bounds on $\text{HJ}(t, r)$ — can coherence be replaced by a graph-theoretic hypothesis?

Chapter 8

Conclusion

This thesis treats the Hales–Jewett numbers through one organising principle: the symmetry of the line-free colourings realising the lower bounds. Restricting to the symmetric level reduces an exponential search over $[t]^n$ to a polynomial one over the type simplex (Chapter 3); the symmetric colourings coincide with the one-weight colourings, so the hierarchy above them is flat (Theorem 3.5) and the symmetric lower-bound problem is finite and completely specified. The same weight functional converts line-freeness into homothety-avoidance over $\mathbb{Z}$ and yields a uniform Gallai-number lower bound (Theorem 2.15), of which van der Waerden’s theorem is the arithmetic case: weight-independent and tight at $(3, 2)$ (Proposition 2.21), lifted at $(3, 3)$ to $\text{HJ}(3, 3) \geq 16$ from a one-line certificate (Theorem 2.22), and topped by the records $\text{HJ}(3, 3) \geq 22$ and $\text{HJ}(4, 2) \geq 14$ at the head of an explicit ladder of certificates (Remark 3.9). Around this core the thesis charts the variants — bracket ceilings, the interval value $\text{HJ}^{(1)}(3) = 5$, the logarithmic cyclic numbers and the separation of the four line families at $(3, 2)$ — the anatomy and irreducibility of the $(4, 2)$ boundary (Chapter 5), and the identification of the symmetric problem with the weak chromatic threshold of the corner hypergraph, with its Local-Lemma and quasipolynomial consequences (Chapter 7). One methodology runs throughout: reduce each question to a finite object and settle it by a certificate verified in two independent ways.

8.1 Directions for Further Research

The most consequential direction is the passage from the symmetric world to the full one. Bounding $\text{HJ}(t, r)$ from above is a forcing statement over all t^n words — out of computational reach — whereas bounding $\kappa_{\text{sym}}(t, r)$ from above is the same forcing confined to the $O(n^{t-1})$ simplex cells, and is

reachable. The three conjectures — Collapse (Conjecture 4.22), Diagonal-only (Conjecture 4.23), extremality (Conjecture 3.15) — each express this passage, and by the offset formulation (Remark 4.25) even a rigorous upper bound on $\varphi(t, r) = \text{HJ}(t, r) - \kappa_{\text{sym}}(t, r)$ would transport an accessible simplex certificate into the first nontrivial upper bound on a Hales–Jewett number from this circle of ideas. Establishing any such offset — or refuting extremality, for which Corollary 7.32 opens an asymptotic route — is the central open problem.

The concrete frontiers are within sight. Deciding the symmetric simplex instance at $(4, 2)$, $n = 14$ would either pin $\text{HJ}(4, 2) = 14$ under extremality or break the record through a radix weight; likewise $(3, 3)$ at $n = 22$. The boundary analysis sharpens the obstacle: the level instance is globally irreducible across support, width, and $\text{GF}(2)$ Nullstellensatz degree (open already at degree 6), yet three monochromatic copies from feasible, so progress likely demands a symmetry-aware refutation that the unquotiented measures cannot see. Past the symmetric reach, the genuine hierarchy is the block ladder of Section 3.4; climbing it, and seeking asymmetric line-free colourings off the simplex, is the natural route beyond the records.

On the one-dimensional side, the Gallai optimum $\max_{|S|=t} \lceil (G_r(S) - 1)/D_S \rceil$ is a sharp extremal problem, settled at $(3, 2)$ and open first at $(3, 3)$, where it lies between 16 and $\text{HJ}_{\text{sym}}(3, 3)$: closing that gap would either replace the 253-cell radix table by a one-dimensional certificate or show that the carry homothets, invisible to the grid, are what the closed form cannot spare (Remark 3.6). The anatomy of Chapter 5 indicates where to look: the $(4, 2)$ champions are single affine orbits whose reach is one boundary scale above the periodic ceiling, while at $(3, 3)$ nine scales are missing — the question is whether quasi-periodic palettes with boundary defects, of which the length-76 certificate (agreeing with its own shift by 19 in 55 of 57 overlapping positions) is a first specimen, can close a many-scale gap.

Among the variants, it is open whether $\text{HJ}^{\text{cyc}}(p, r) = O_p(\log r)$ persists for primes $p \geq 5$, where a full line is a p -term progression beyond the three-term relation the polynomial method controls; and, with the interval ceiling $\lambda(3, \cdot)$ now determined (Proposition 4.19), the finite interval numbers beyond $\text{HJ}^{(1)}(3) = 5$ — the first unknown being $\text{HJ}^{(3)}(3, 3)$ — remain to be charted (Question 4.21). Structurally, Question 7.35 asks for an intrinsic, coherence-free parameter of the intersection graph equivalent to bounds on HJ. Finally, on the anti-Ramsey side — where symmetric colourings are tight for the monochromatic records but exponentially lossy — the rainbow analogue calls for a Combinatorial Nullstellensatz in place of the all-equal ideal, and the value $\text{ah}(3, 4) \in [25, 27]$ (Proposition 5.9) remains to be closed.

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Appendix A

Certificates and Tables

This appendix reproduces the certificates and data tables underlying the computer-assisted results of the thesis. Each avoidance certificate — a palette, simplex table, or interval colouring — is a complete proof of its lower bound: the verification is a finite check against the definitions (Lemmas 2.14 and 3.1), independent of any solver, and has been carried out by the standalone, dependency-free verifiers of the public repository

<https://github.com/ysmouhib/hj-certificates>

which also contains the SAT encoders, the deletion-filter and MaxSAT drivers, the solver logs for every unsatisfiability (forcing) claim, the census data of Section A.4, and the remaining machine-readable certificates not reproduced here. Palettes printed inline in the main text (the length-76 interval certificate of Theorem 2.22, the 49-cell palette of Proposition 3.8, the period-26 record palette of Theorem 3.10, and the mod-13 palettes of Theorems 4.10 and 4.11) are not repeated.

A.1 Record Certificates

$HJ(3, 3) \geq 22$. Table A.1 is the witness of Theorem 3.7: a 3-colouring of the 253 cells of the simplex $T_{21}^{(3)}$ leaving none of the 1771 corner triples $C_{k,v}$ ($1 \leq k \leq 21$) monochromatic. Row a lists the colours of the cells with a ones; the character at position b ($b = 0, \dots, 21 - a$) is the colour of the cell $(a, b, 21 - a - b)$, i.e. of the words with a ones, b twos and $21 - a - b$ threes. The verifier re-derives all 1771 triples from the definition and reports zero monochromatic; the count $1771 = \binom{23}{3}$ certifies that no tuple was skipped (Remark 3.2).

Table A.1: A 3-colouring of $T_{21}^{(3)}$ with no monochromatic corner triple, certifying $\text{HJ}(3, 3) \geq 22$. Row a , position b : colour of the cell $(a, b, 21-a-b)$.

a	colours at $b = 0, 1, \dots, 21 - a$
0	0102210210210210210022
1	121002102102102102210
2	20211021021021021102
3	2210121001102022210
4	002111211020021102
5	11022202100210021
6	2210102022021210
7	002120011211102
8	11020112202021
9	2210021102210
10	022111020102
11	01020110021
12	1210002211
13	202110102
14	01022021
15	1210210
16	202202
17	01101
18	1001
19	120
20	20
21	0

$\text{HJ}(4, 2) \geq 14$. The certificate is the period-26 palette printed in Theorem 3.10; its verification enumerates the 253 residue tuples $(b, b+2k, b+3k, b+5k) \bmod 26$ realised on $[4]^{13}$ and, for the sharpness and anatomy statements (Proposition 5.3), the full 26×14 base-scale table. The equivalent simplex form — a 2-colouring of the 560 cells of $T_{13}^{(4)}$ avoiding all $1820 = \binom{16}{4}$ corner quadruples — is in the repository.

A.2 Gallai and Arithmetic Certificates

The window certificate for $(0, 1, 3)$ at four colours. The following 4-colouring χ of $\{0, \dots, 92\}$ has no monochromatic homothet $\{b, b+k, b+3k\}$, $k \geq 1$, verified exhaustively over all such triples in the window; it establishes $G_4(\{0, 1, 3\}) \geq$

94 and the grid bound of Remark 2.24. It is *not* solution-free for $z + 2x = 3y$: the injective solutions of the equation also comprise the reflected copies $b + k\{0, 2, 3\}$ (Example 2.20), of which χ contains monochromatic instances; the Rado bound rests on the separate certificate below.

$\chi = (2, 2, 1, 3, 0, 2, 2, 0, 1, 1, 1, 3, 3, 1, 2, 0, 3, 1, 1, 0, 0, 3, 3, 3, 1, 0, 2, 2, 0, 3, 1, 2, 0, 2, 2, 0, 1, 2, 0, 1, 0, 0, 1, 1, 1, 3, 0, 1, 3, 0, 3, 3, 3, 2, 2, 3, 1, 2, 1, 0, 2, 1, 2, 0, 1, 1, 1, 3, 3, 2, 0, 2, 1, 0, 0, 1, 3, 3, 1, 2, 0, 2, 3, 0, 2, 3, 3, 1, 0, 3, 3, 1)$.

A solution-free certificate for $z + 2x = 3y$ at four colours. The following 4-colouring of 56 consecutive integers contains no monochromatic injective solution of $z + 2x = 3y$ — equivalently (Example 2.20), no monochromatic homothet of $\{0, 1, 3\}$ or of $\{0, 2, 3\}$ — verified exhaustively against all solutions of the equation in the window. Since the predicate is translation-invariant, it establishes $R_4(z + 2x = 3y) \geq 57$ (Remark 2.24).

32013003130311321322210002211032
220120031301113013223120

(listing $\chi(0), \dots, \chi(55)$).

The power-residue palette for $\kappa_{\text{sum}}(4, 3) = 96$. The construction of Corollary 4.8 at $p = 97$ uses the primitive root $g = 5$ and the 97-periodic palette $\chi(x) = \text{ind}_g(x) \bmod 3$ ($\chi(0) = 0$), explicitly

0011212102220122120002021220010120021112110110002
200011011211120021010022120200021221022201212110

(listing $\chi(0), \dots, \chi(96)$). Direct enumeration of all 97×96 progressions confirms no monochromatic 4-term arithmetic progression of gap 1, $\dots$, 96, while gap 97 collapses modulo p and is monochromatic: the construction ceiling is exactly 96, matching the upper bound $\lfloor (W(4, 3) - 1)/3 \rfloor - 1 = 96$ from $W(4, 3) = 293$.

The 12-periodic palette for $\kappa_{\text{sum}}(3, 3) = 11$. The palette $(2, 0, 1, 2, 1, 1, 0, 1, 2, 0, 0, 2)$ of Proposition 4.5 carries no monochromatic 3-term progression of gap 1, $\dots$, 11; the check runs over the 12×11 residue pairs.

Remaining Gallai certificates. The avoidance colourings realising $G_3(\{0, 1, 3\}) > 41$, $G_3(\{0, 1, 4\}) > 56$, $G_2(\{0, 2, 3, 5\}) > 66$ and $G_2(\{0, 1, 5, 6\}) > 79$ of Table 2.1, and the fourteen two-colour values verifying Proposition 2.21, are in the repository together with the solver logs of the matching forcing runs; the length-76 certificate for $G_3(\{0, 2, 5\}) \geq 77$ is printed in Theorem 2.22.

A.3 The Interval Number $HJ^{(1)}(3) = 5$

Theorem 4.20 rests on two satisfiability computations, each re-checked against the definition without trusting the solver.

Lower bound $HJ^{(1)}(3) \geq 5$. The instance asking whether $[3]^4$ admits a 2-colouring with no monochromatic line in $L^{(1)}([3]^4)$ — no monochromatic line whose active coordinates form a single contiguous block — is satisfiable, and the decoded witness is displayed in Table A.2; it was re-checked, independently of the solver, against all 142 lines of $L^{(1)}([3]^4)$ by direct enumeration from the definition, with zero monochromatic lines found.

Table A.2: A 2-colouring of $[3]^4$ with no monochromatic line in $L^{(1)}([3]^4)$. The entry in row (x_1, x_2) and column (x_3, x_4) is the colour of the word (x_1, x_2, x_3, x_4) .

x_1	x_2	(1,1)	(1,2)	(1,3)	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
1	1	1	1	0	1	0	0	0	1	1
1	2	0	1	0	1	0	1	1	0	1
1	3	1	0	1	0	1	1	0	1	0
2	1	0	1	0	0	1	1	1	0	0
2	2	1	0	1	0	1	1	0	1	0
2	3	0	1	1	1	0	0	1	0	1
3	1	1	0	1	1	0	0	0	1	1
3	2	0	1	1	0	1	0	1	0	1
3	3	0	1	0	0	1	1	1	0	0

Upper bound $HJ^{(1)}(3) \leq 5$. The same encoding one dimension higher, for $[3]^5$, ranges over all 2^{3^5} colourings — no symmetry is assumed — and is *unsatisfiable*: every 2-colouring of $[3]^5$ contains a monochromatic line in $L^{(1)}([3]^5)$. To make the upper bound auditable, a deletion filter extracts a *forcing subfamily*: a minimal set $F \subseteq L^{(1)}([3]^5)$ of lines such that no 2-colouring of the words occurring in F avoids a monochromatic member of F , while every proper subset of F does admit one. The family found consists of 262 lines on 172 of the 243 words of

$[3]^5$; its distribution over the active intervals is given in Table A.3, and the lines themselves are reproduced and re-verified in the repository. The certificate is large: unlike $\text{HJ}^{(1)}(2, 3) = 3$, which is forced by a four-word K_4 (Example 4.16), no small forcing structure exists here — across several elimination orders, the minimal subfamilies all numbered in the hundreds of lines.

Table A.3: Distribution of the 262-line forcing subfamily over the active intervals $[i, j]$ of $[5]$. Each cell shows the number of family lines with that active interval, out of the $3^{5-(j-i+1)}$ lines of $L^{(1)}([3]^5)$ available; the entries sum to 262/547.

start i	end j				
	1	2	3	4	5
1	5/81	14/27	8/9	3/3	1/1
2		27/81	16/27	6/9	2/3
3			50/81	17/27	8/9
4				59/81	17/27
5					29/81

A.4 Diagonal-Only Witnesses and the $[3]^3$ Censuses

A.4.1 A 2-Coloring of $[3]^4$ Whose Unique Monochromatic Line Is the Diagonal

The satisfiable instance $(t, n, r) = (3, 4, 2)$ decodes to the colouring of Table A.4. It was re-checked, independently of the solver, against all 175 combinatorial lines of $[3]^4$ by direct enumeration: exactly one line is monochromatic, namely the main diagonal $L_{***} = \{(1, 1, 1, 1), (2, 2, 2, 2), (3, 3, 3, 3)\}$ (entries in bold, all of colour 0). In particular no line of $L^{[3]}([3]^4)$ is monochromatic, so $\text{HJ}^{[3]}(3) \geq 5$, as used in Section 4.5.

A.4. Diagonal-Only Witnesses and the $[3]^3$ Censuses

Table A.4: A 2-colouring of $[3]^4$ whose unique monochromatic line is the main diagonal. The entry in row (x_1, x_2) and column (x_3, x_4) is the colour of the word (x_1, x_2, x_3, x_4) .

x_1	x_2	(1,1)	(1,2)	(1,3)	(2,1)	(2,2)	(2,3)	(3,1)	(3,2)	(3,3)
1	1	0	1	0	1	1	0	0	0	1
1	2	1	1	0	1	0	1	0	1	0
1	3	0	0	1	0	1	0	1	0	1
2	1	1	1	0	1	0	1	0	1	0
2	2	1	0	1	0	0	1	1	1	0
2	3	0	1	0	1	1	0	0	0	1
3	1	0	0	1	0	1	0	1	0	1
3	2	0	1	0	1	1	0	0	0	1
3	3	1	0	1	0	0	1	1	1	0

A.4.2 A Census of the Line-Free 2-Colorings of $[3]^3$

Since $\text{HJ}(3, 2) = 4$ [2], the grid $[3]^3$ is the largest in which a line-free 2-colouring exists, so its colourings admit a complete account. There are exactly 1644 of them. They split first by the coordinate-permutation stabilizer — the invariance measured at the top of the hierarchy $\text{sum} \subseteq \text{ow} \subseteq \text{sym}$ of Section 2.2, and by its proper subgroups below — and the symmetric ones split further by the weight that realises them. The counts are exhaustive: every one of the 2^{27} colourings was tested against the 37 combinatorial lines, the symmetric total was cross-checked on the ten-cell simplex $T_3^{(3)}$ through its ten corner triples (Lemma 3.1), and the sum-type total against the nine weight-progressions $S + k\{1, 2, 3\} \subseteq [3, 9]$.

A.4. Diagonal-Only Witnesses and the $[3]^3$ Censuses

Table A.5: The 1644 line-free 2-colourings of $[3]^3$, by coordinate symmetry and by weight.

class	defining invariance / weight [*]	number	orbits [†]
symmetric (= one-weight) ^a	coordinate stabilizer S_3	36	36
sum-type	weight (1, 2, 3)	16	16
one-weight, non-arithmetic ^a	a non-arithmetic weight	20	20
non-symmetric	coordinate stabilizer $\subsetneq S_3$	1608	360
block-symmetric ^b	stabilizer C_2 (one transposition)	504	168
cyclic	stabilizer C_3 (the 3-cycles)	24	12
asymmetric	trivial stabilizer	1080	180
total		1644	396

^{*} The four classes partition the colourings by their stabilizer in the coordinate group S_3 (whose subgroups are 1, C_2 , C_3 , S_3); the two indented rows under *symmetric* instead subdivide that class by the weight ω realising it (Definition 2.6).

[†] Orbits under the coordinate action of S_3 ; by orbit–stabilizer each orbit has size $6/|\text{stab}|$. Under $S_3^{\text{coord}} \times S_3^{\text{alph}}$ there are 66 orbits, and 33 under that group together with the colour swap. The stabilizer under $S_3^{\text{coord}} \times S_3^{\text{alph}}$ equals the coordinate stabilizer for *every* one of the 1644 colourings — none gains symmetry from the alphabet, in accordance with Proposition 3.4.

^a In a single fixed dimension an injective weight on $T_3^{(3)}$ (e.g. $\omega = (0, 1, 4)$, giving the ten distinct values 0, 1, 2, 3, 4, 5, 6, 8, 9, 12) realises *any* symmetric descent, so $\text{ow} = \text{sym}$ here; the strict chain $\text{sum} \subsetneq \text{ow} \subsetneq \text{sym}$ of Remark 2.7 is an all-dimensions statement. The 16/20 split is nonetheless intrinsic: each of the 20 is realised by a non-injective *non-arithmetic* weight (e.g. (0, 2, 3) or (0, 1, 3)) and by no arithmetic weight.

^b Block-symmetric (Section 3.4) for a partition with a block of size ≥ 2 ; for $n = 3$ this forces invariance under exactly one transposition, two transpositions generating S_3 and returning the colouring to the symmetric class.

Remark A.1. The intermediate tiers are genuinely populated at $[3]^3$: twenty symmetric colourings are one-weight but not sum-type, and twenty-four are cyclic — invariant under the 3-cycles yet under no transposition, hence *not* block-symmetric, the group C_3 being no Young subgroup. The remaining 1080 carry no coordinate symmetry at all. Every class is closed under the colour swap, so the 1644 colourings form 822 complementary pairs and the 36 symmetric ones 18.

A.4.3 A Census of the $[3]^3$ Colorings Whose Unique Monochromatic Line Is the Diagonal

The $[3]^4$ witness of Table A.4 has an $n = 3$ analogue, and there the family can be described completely. Call a 2-colouring of $[3]^3$ *diagonal-only* if the main diagonal $L_{***} = \{(1, 1, 1), (2, 2, 2), (3, 3, 3)\}$ is monochromatic and every

other one of the 37 combinatorial lines is bichromatic. Since the diagonal is the unique line with all three coordinates active, a diagonal-only colouring has no monochromatic line in $L^{[2]}([3]^3)$; it is the $n = 3$ counterpart of Table A.4, which at $n = 4$ gives $\text{HJ}^{[3]}(3) \geq 5$. At $n = 3$, however, line-free colourings already exist (the 1644 of Table A.5), so the diagonal-only family is a refinement of that census rather than a lower-bound certificate: it isolates the colourings whose sole obstruction is the top, $k = 3$, line.

There are exactly 6456 of them — 3228 with the diagonal of colour 0 and 3228 with it of colour 1, interchanged by the colour swap with no fixed point, hence 3228 complementary pairs. Table A.6 gives the classification by coordinate symmetry, alongside the line-free census for comparison. All counts are exhaustive: each of the 2^{27} colourings was tested against the 37 lines; the symmetric total was cross-checked on the simplex $T_3^{(3)}$, where by Lemma 3.1 a diagonal-only symmetric colouring is exactly a $\bar{c}: T_3^{(3)} \rightarrow \{0, 1\}$ making the corner triple $C_{3,0} = \{(3, 0, 0), (0, 3, 0), (0, 0, 3)\}$ monochromatic and the other nine non-monochromatic; and the sum-type total against the nine progressions $S + k\{0, 1, 2\} \subseteq [0, 6]$.

A.4. Diagonal-Only Witnesses and the $[3]^3$ Censuses

Table A.6: The 6456 diagonal-only 2-colourings of $[3]^3$, by coordinate symmetry and by weight; the line-free counts of Table A.5 are shown for comparison.

class	defining invariance / weight [*]	number	orbits [†]	line-free [‡]
symmetric (= one-weight) ^a	coordinate stabilizer S_3	34	34	36
sum-type	weight (1, 2, 3)	14	14	16
one-weight, non-arithmetic ^a	a non-arithmetic weight	20	20	20
non-symmetric	coordinate stabilizer $\subsetneq S_3$	6422	1296	1608
block-symmetric ^b	stabilizer C_2 (one transposition)	1338	446	504
cyclic	stabilizer C_3 (the 3-cycles)	8	4	24
asymmetric	trivial stabilizer	5076	846	1080
total		6456	1330	1644

^{*} The four classes partition the colourings by their stabilizer in the coordinate group S_3 (whose subgroups are 1, C_2 , C_3 , S_3); the indented rows under *symmetric* subdivide that class by the weight ω realising it (Definition 2.6).

[†] Orbits under the coordinate action of S_3 ; each orbit has size $6/|\text{stab}|$. Up to $S_3^{\text{coord}} \times S_3^{\text{alph}}$ there are 254 orbits, and 127 under that group together with the colour swap. As for the line-free census, the stabilizer under $S_3^{\text{coord}} \times S_3^{\text{alph}}$ equals the coordinate stabilizer for every diagonal-only colouring (Proposition 3.4).

[‡] The corresponding line-free counts of Table A.5.

^a As in Remark 2.7, $\text{ow} = \text{sym}$ in fixed dimension (an injective weight on $T_3^{(3)}$ realises any symmetric descent), so the literal count of one-weight colourings is 34; the 14/20 split is intrinsic, each of the 20 being realised by a non-injective *non-arithmetic* weight (e.g. (0, 2, 3)) and by no arithmetic weight.

^b Block-symmetric (Section 3.4) for a partition with a block of size ≥ 2 ; for $n = 3$ this is invariance under exactly one transposition.

Remark A.2. Forcing the diagonal monochromatic is invariant under both the coordinate and the alphabet action, yet it ties the remaining 24 words only loosely, so the mass moves into the low-symmetry strata: against the line-free census the block-symmetric and asymmetric classes grow ($504 \rightarrow 1338$, $1080 \rightarrow 5076$) while the cyclic class shrinks ($24 \rightarrow 8$) and the non-arithmetic one-weight count is unchanged at 20. The symmetric total moves $36 \rightarrow 34$ and the sum-type total $16 \rightarrow 14$, the two-colouring difference accounted for on the simplex by trading the requirement that $C_{3,0}$ be bichromatic for the requirement that it be monochromatic.



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